

CO 342 Intro to Graph Theory

Cuiwen Zhu

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1 Connectivity

G is 1-connected is same as G is connected if $|V(G)| \geq 2$, otherwise not.

Definition 1.1. We write $\delta(G)$ to be the minimum degree of G .

Remark 1.2. 1. If G is k -connected, then $\delta(G) \geq k$.

2. If $\delta(G) \geq \frac{|V(G)|+k-2}{2}$, then G is k -connected.

1.1 Properties of 2-connected graphs

Lemma 1.3. If G is a 2-connected graph and $xy, xz \in E(G)$, then there is a cycle in G containing both xy and xz .

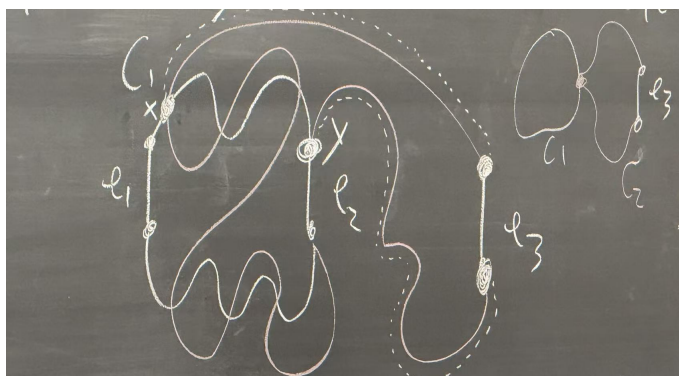
Proof. Since G is connected, x is not a cutvertex. So $G - x$ is connected. Hence there exists a path P from y to z in $G - x$, then $P \cup \{x\}$ forms a cycle in G containing xy and xz . $\square$

Lemma 1.4. Let G be a graph. Suppose that

- edges e_1 and e_2 lie in a common cycle in G , and
- edges e_2 and e_3 lie in a common cycle in G ,

then e_1 and e_3 lie in a common cycle in G .

Proof. Let e_1, e_2 be in cycle C_1 , and e_2, e_3 be in cycle C_2 . Let x be the first vertex of C_1 reached by walks along C_2 starting from one endpoint of e_3 and not using e_3 . Similarly, let y be the first vertex of C_1 reached by walks along C_2 starting from the other endpoint of e_3 and not using e_3 . Notice that x, y exist and $x \neq y$ since e_2 is in both C_1 and C_2 , so the “worst” case is x, y are the two endpoints of e_2 . Then the (x, y) -segment of C_2 contains e_3 , and the (y, x) -segment of C_1 contains e_1 , so we have a cycle containing e_1 and e_3 .



$\square$

1.2 Alternate Characterizations of 2-connectivity

Theorem 1.5. Let G be a graph with $|V(G)| \geq 3$. TFAE

- (a) G is 2-connected.
- (b) G has no isolated vertices and every pair of edges lies in a common cycle.
- (c) Any two vertices lie in a common cycle.

Proof. (a) $\implies$ (b): Suppose G is 2-connected, then G has no isolated vertices since $\delta(G) \geq 2$. Let e and f be two edges in G . Since G is connected, there is a path from an endpoint of e to an endpoint of f . Then there is a sequence of edges $e_0, e_1, \dots, e_t$ where $e_0 = e, e_t = f$, and e_i shares a vertex with e_{i+1} for each i . By applying our first lemma, for each i , there is a cycle C_i containing e_i and e_{i+1} . By applying our second lemma inductively, $e_0 = e$ and $e_t = f$ are in the same cycle.

(b) $\implies$ (c): Let x, y be a pair of arbitrary vertices. Since there is no isolated point in G , x, y are not isolated, so there exists e containing x and f containing y , with at least one of e, f is not $x - y$ (as $|V(G)| \geq 3$), therefore we can find a common cycle containing e, f , therefore containing x, y .

(c) $\implies$ (a): Suppose, for contradiction, that x is a cut vertex of G . Let y, z be vertices in distinct components of $G - x$, then there is no common cycle C containing y, z , otherwise removing x cannot disconnect y and z in C . $\square$

What is a graph like if it is connected but not 2-connected?

1.3 Blocks

Definition 1.6. A block is a connected graph with no cutvertex. So a block is one of the following:

- 1. a single vertex (trivial block)



- 2. the graph K_2



- 3. a 2-connected graph

Definition 1.7. A block of a graph G is a subgraph H of G that is a block, and is maximal with respect to being a block.

Remark 1.8. By the definition, we have

- H is a block, and
- H is not contained in any larger subgraph T of G that is also a block.

Example 1.9. G has 8 blocks:

Theorem 1.13. Let G be a connected graph. The BCF of G is a tree.

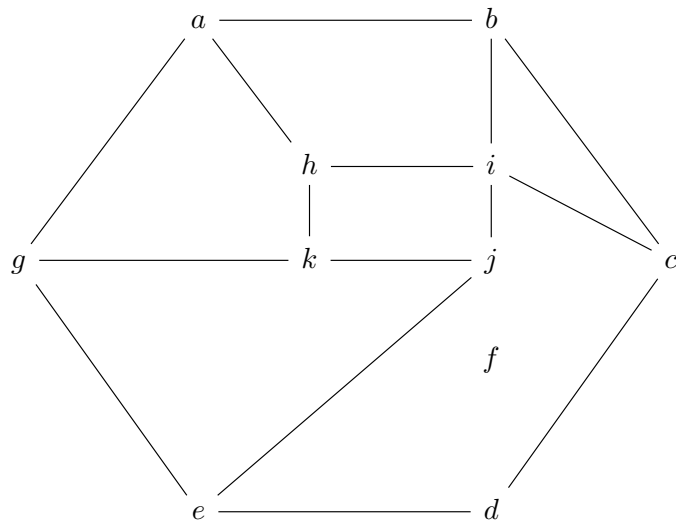
Proof. Suppose, for contradiction, that the BCF of G contains cycles. Let $c_1b_1 \dots c_tb_t$ be a shortest cycle in BCF of G . Then $t \geq 2$. Also, $b_i - \{c_{i-1}, c_i\}$ are disjoint since we chose a shortest cycle in BCF. Then there is a path P_i inside b_i from c_{i-1} to c_i for each i , and these are all disjoint except for c_i 's. Hence $P = P_3 \cup P_4 \cup \dots \cup P_t$ is a path from c_2 to c_t that does not contain c_1 . Hence, as in the proof of previous lemma, $b_1 \cup b_2 \cup P$ is 2-connected, which violates the maximality of b_1 . Therefore, BCF is a tree. $\square$

Definition 1.14. An end-block of a connected graph G is a block of G that has degree 1 in the BCF of G (that is, a leaf of BCF).

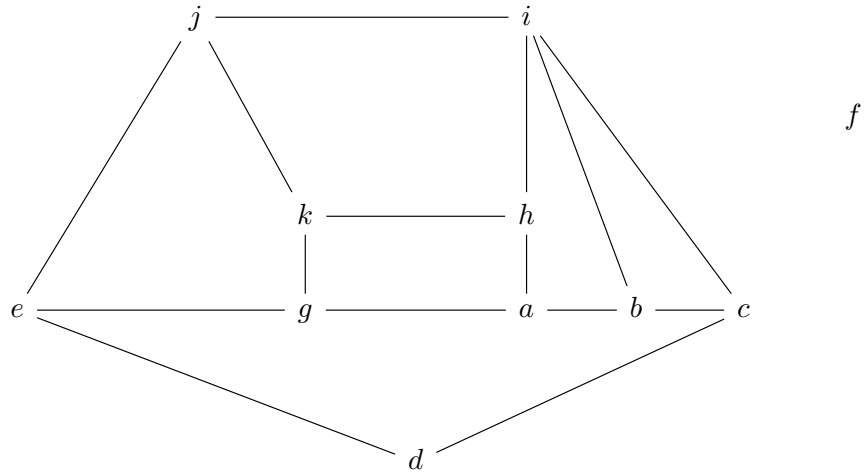
Remark 1.15. Notice that every leaf of the BCF is a block, since any cutvertex separates the graph so is in at least 2 blocks, which guarantee the existence of end-blocks for graph with at least 2 blocks.

Recall: For any planar graph G , if $\tilde{G}$ is a planar drawing of G and f is a face of $\tilde{G}$, then there exists a planar drawings $\tilde{G}_1$ of G in which f is the outer face.

Example 1.16. We have



and



Theorem 1.17 (K1). Let G be a connected graph, and suppose every block of G is planar. Then G is planar.

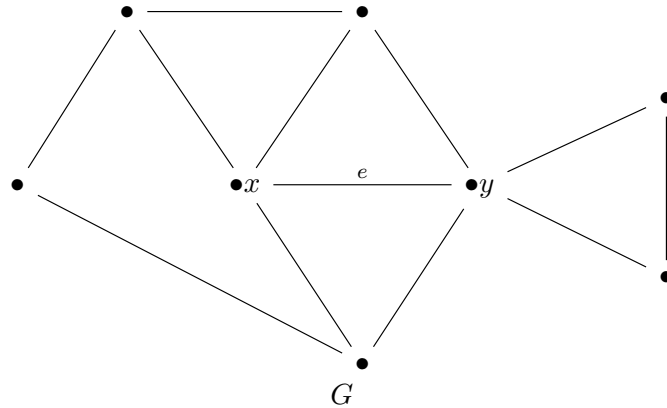
Proof. We use induction on the number t of blocks of G .

Base case: $t = 1$. Then G is itself a block. Hence G is planar by assumption.

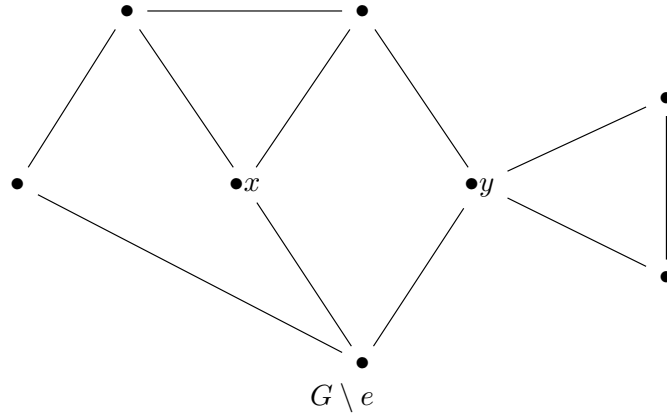
Inductive Hypothesis: Assume $t \geq 2$ and every connected graph with $\leq t - 1$ blocks, all of which are planar.

Let G with t blocks be given. Since $t \geq 2$, G has an end-block b . Let c be the unique cutvertex of G in b . By IH, the graph $H = (G - b) \cup \{c\}$ is planar, since it has $t - 1$ blocks, all of which were blocks of G and hence planar by assumption. Also, b is planar since it is a block of G . We take a planar drawing of H with c on the outer face (by lemma), and a planar drawing of H with b on the outer face. $\square$

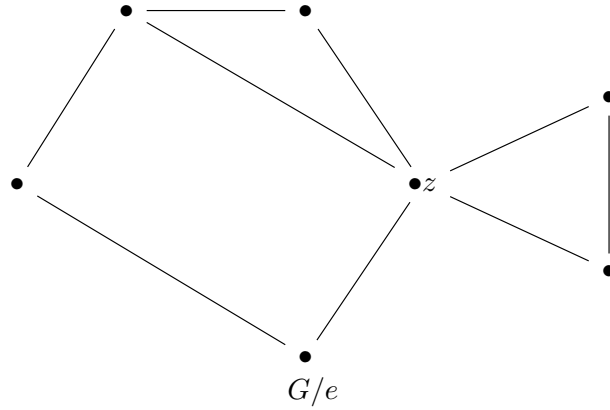
Definition 1.18 (Contraction and Deletion of edges). Let G be a graph and let $e = xy$ be an edge of G .



Then a deletion $G \setminus e$ is “erasing” the edge e :



And a contraction G/e is “gluing” e to be a new vertex z :



Formally, the graph $G \setminus e$ obtained by deleting e from G has $V(G \setminus e) = V(G)$ and $E(G \setminus e) = E(G) \setminus \{e\}$.

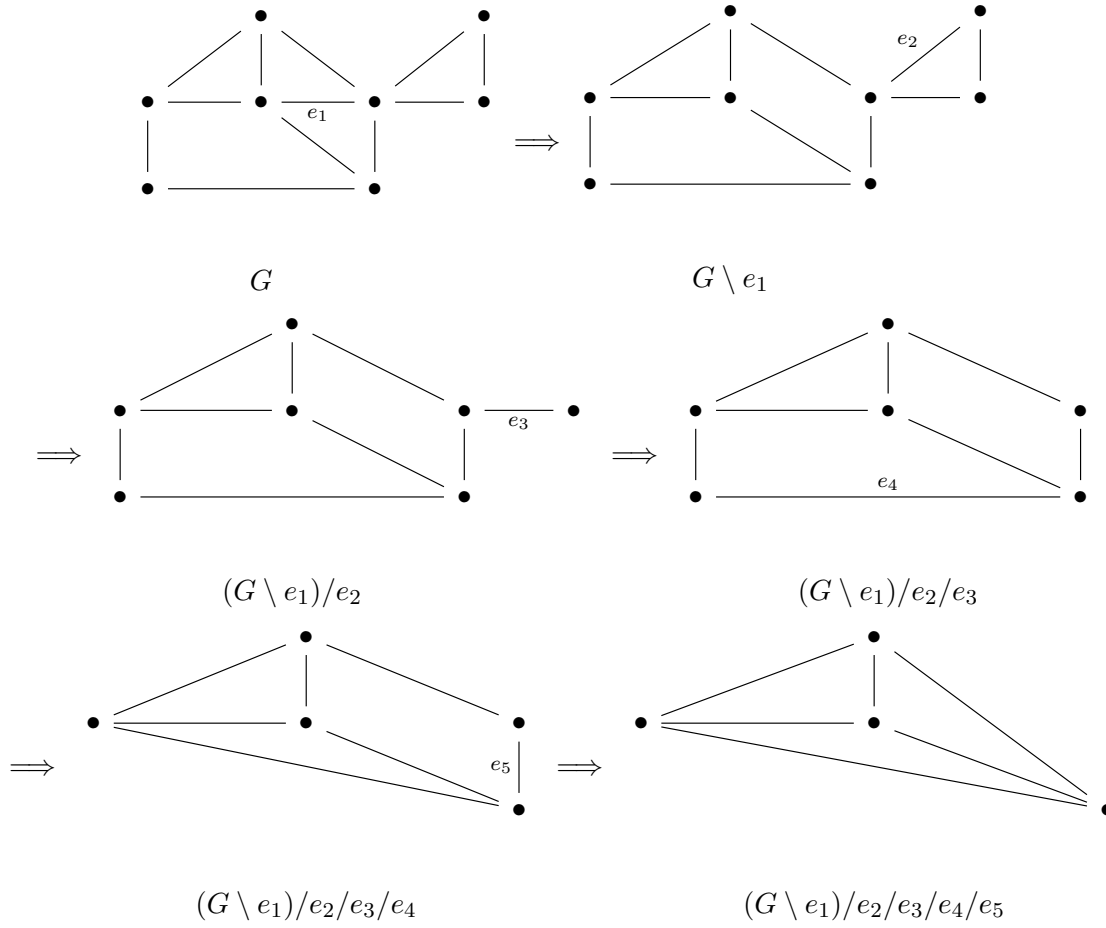
The graph G/e obtained by contracting e in G has $V(G/e) = (V(G) \setminus \{x, y\}) \cup \{z\}$, and

$$E(G/e) = \{uv \in E(G) : \{u, v\} \cap \{x, y\} = \emptyset\} \cup \{uz : ux \in E(G) \text{ or } uy \in E(G) \text{ (or both)}\}$$

Remark 1.19. Note that $|E(G \setminus e)| = |E(G)| - 1$, and $|E(G/e)| \leq |E(G)| - 1$. So they are good for inductions.

Definition 1.20. The graph H is said to be a minor of the graph G if H can be obtained from G by a sequence of edge deletions and edge contractions (plus deletion of isolated vertices).

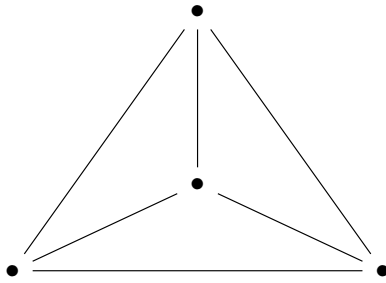
Example 1.21. We show that K_4 is a minor of the following G via a sequence of deletion and contraction:



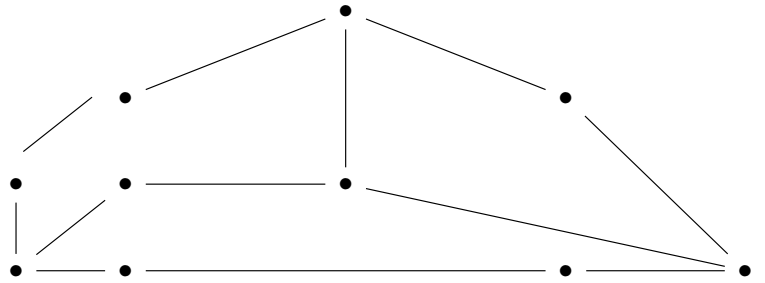
Definition 1.22. Let H be a graph. A subdivision J of H is any graph J obtained from H by replacing each edge e of H by a path P_e of length ≥ 1 (whose endpoints are the endpoints of e) such that

- $V(P_e) \cap V(P_f) = \{x\}$ whenever $e \cap f = \{x\}$,
- $V(P_e) \cap V(P_f) = \emptyset$ whenever $e \cap f = \emptyset$.

Example 1.23.



H



a subdivision of H

Remark 1.24. Note that

- $d_J(v) = d_H(v)$ if $v \in V(J) \cap V(H)$ “branch” vertices of J ,
- $d_J(v) = 2$ if $v \in V(J) \setminus V(H)$ “path” vertices of J .

Also, a subdivision of a subdivision is a subdivision.

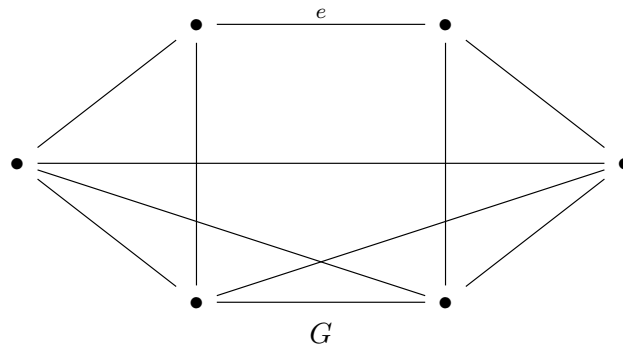
Now let’s compare the two statements:

(a) G has H as a minor

(b) G contains a subdivision of H .

(b) $\implies$ (a): Suppose G contains a subdivision J of H . Start a sequence by deleting all edges of G not in J , to obtain J (plus isolated vertices – delete these), then contract each P_e in J down to a single edge to get H .

(a) $\not\implies$ (b) in general:



Then G/e is K_5 , therefore G contains K_5 as a minor. However, G does not contain a subdivision J of K_5 since J would need 5 vertices of degree 4 (the branch vertices) and G has only 4 vertices of degree 4.

Definition 1.25. A cubic graph G is a graph in which every vertex has degree 3 (same as 3-regular).

Lemma 1.26. If H is a cubic graph, then (a) $\iff$ (b). That is, if a graph G has H as a minor, then G contains a subdivision of H .

Proof. We will prove this lemma by induction on $|E(G)|$.

Base case: $|E(G)| = |E(H)|$. Then G (after deleting the empty vertices) equals to H , and since H is a subdivision of itself, the statement is true.

IH: Assume $|E(G)| > |E(H)|$, and for every graph G' with $|E(G')| \leq |E(G)| - 1$, that contains H as a minor, contains a subdivision of H .

Fix a sequence S of edge deletions and contractions that takes G to H . Then we can split into two cases:

1. The first step of S is a deletion of edge e . Then $G \setminus e$ satisfies $|E(G \setminus e)| = |E(G)| - 1$, and the sequence S except for its first step brings $G \setminus e$ to H , that is, $G \setminus e$ contains H as a minor. By IH, we know that $G \setminus e$ contains a subdivision J of H . Hence G contains a subdivision J as well.
2. The first step of S is a contraction of edge e . Let $e = xy$ and it contracts to z . As before, the sequence S without the first step shows G/e has H as a minor. Since $|E(G/e)| < |E(G)|$, by IH, G/e contains a subdivision H_1 of H . If $z \notin V(H_1)$, then H_1 is not affected by the contraction and therefore H_1 is a subgraph of G . So G contains a subdivision of H as desired. If $z \in V(H_1)$, since H is cubic, we then have $d_{H_1}(z) = 2$ if it is a path vertex of H_1 and $d_{H_1} = 3$ if it is a branch vertex of H_1 . Let H_2 be the minimal subgraph of G such that $H_2/e = H_1$. In particular, $d_{H_1}(z) \leq 3$. Thus, of the ≤ 3 edges that z incidents in H , one of x and y is incident to at most one of them in H_2 . It follows that H_2 contains a subdivision of H (possibly with one path length extended by 1). Hence G contains a subdivision of H .

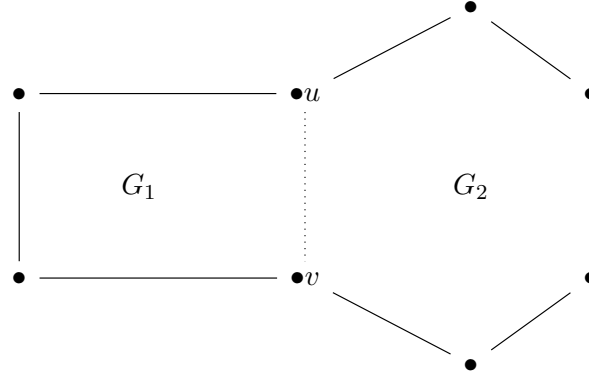
□

Definition 1.27. Let G_1 and G_2 be graphs with at least 3 vertices. Suppose that

- $V(G_1) \cap V(G_2) = \{u, v\}$ where $u \neq v$,
- $uv \in E(G_1) \cap E(G_2)$.

Then the 2-sum $G_1 \oplus G_2$ of G_1 and G_2 is the graph with vertex set $V(G_1) \cup V(G_2)$ and edge set $E(G_1) \cup E(G_2) \setminus \{uv\}$

Example 1.28. We can have



$$G_1 \oplus G_2$$

Remark 1.29. Any 2-sum of two cycles is a cycle. In the example above, $C_4 \oplus C_6 = C_8$.

Lemma 1.30. If $G = G_1 \oplus G_2$ via the edge uv , and if G_2 is 2-connected, then G contains a subdivision of G_1 .

Proof. Since G_2 is 2-connected, there exists a cycle C in G_2 containing uv (by the characterization of 2-connected graphs (3)). Then $C \setminus uv$ is a path in G from u to v that is disjoint from $V(G_1) \setminus \{u, v\}$. Hence, $C \setminus uv$ together with the copy of $G_1 \setminus uv$ in G gives a subdivision of G_1 . $\square$

Lemma 1.31. If $G = G_1 \oplus G_2$ via uv , and G_1, G_2 are 2-connected. Then G is 2-connected.

Proof. We will prove it by proving G has no isolated vertices and every pair of edges lies in a common cycle.

- G has no isolated vertices since $\delta(G_1) \geq 2$ and $\delta(G_2) \geq 2$, and at most one edges is removed from G .
- Every pair of edges e, f in G lies in a common cycle:
 1. If $e, f \in E(G_1)$ (or $E(G_2)$), since G_1 (or G_2) is 2-connected, there exists a cycle C containing e and f . If $uv \notin E(C)$, then we are done. If $uv \in E(C)$, then let C' be a cycle in G_2 containing uv , then $C = C_1 \oplus C_2$ is a cycle in G contains e and f .
 2. If $e \in E(G_1)$ and $f \in E(G_2)$. Since $uv \in E(G_1) \cap E(G_2)$, then there exists cycle C_1 containing uv and e , and a cycle containing uv and f . Therefore $C = C_1 \oplus C_2$ is a cycle containing e and f .

$\square$

Lemma 1.32. Let G be a 2-connected graph and suppose $\{x, y\}$ is a vertex cut of G . Let G be a component of $G - \{x, y\}$. Then the graph H with vertex set $V(G_1) \cup \{x, y\}$ and edge set $\{wz \in E(G) : w, z \in V(H)\} \cup \{xy\}$ is 2-connected.

Proof. $|V(H)| \geq 3$ since $\{x, y\} \subset V(H)$ and C_1 is non-empty. Also, H is connected since C_1 and G are connected. Suppose, for contradiction, that w is a cutvertex of H . Then we may assume $w \neq y$. Let K be a component of $H - w$ that does not contain y . Then w is a cutvertex of G that separates K from the rest. Contradicts to G is 2-connected. Therefore, H is 2-connected. $\square$

Lemma 1.33. Let G_1, G_2 be k -connected graphs, where $|V(G_1) \cap V(G_2)| \geq k$. Then $G = G_1 \cup G_2$ (union of graphs is the union of vertices and edges) is k -connected.

Proof. Clearly $|V(G)| \geq |V(G_1)| \geq k + 1$. Let W be a set of at most $k - 1$ vertices. Then:

- $G_1 - W$ is connected since G_1 is k -connected,
- $G_2 - W$ is connected since G_2 is k -connected,
- $V(G_1) \cap V(G_2) \setminus W$ is non-empty since $|W| < |V(G_1) \cap V(G_2)|$

So $G_1 \cup G_2 - W$ is connected. Hence G has no vertex cut of size $\leq k - 1$, as desired. $\square$

Theorem 1.34 (Decomposition Theorem). Suppose G is a 2-connected graph and $\{x, y\}$ is a vertex cut of G . Then there exists graphs G_1 and G_2 such that

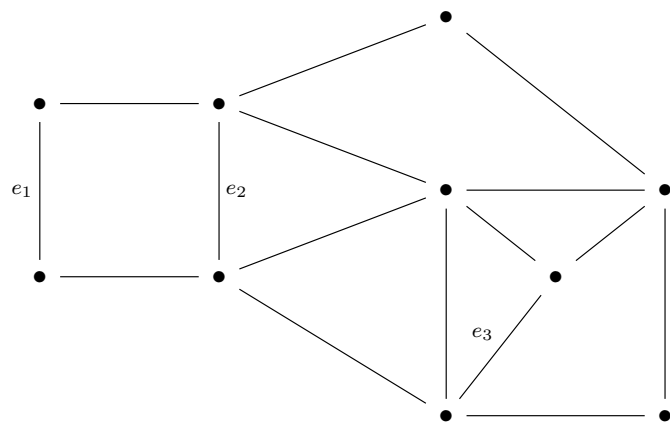
1. $V(G_1) \cap V(G_2) = \{x, y\}$,
2. $xy \in E(G_1) \cap E(G_2)$,
3. G_1, G_2 are 2-connected,
4. if $xy \notin E(G)$, then $G = G_1 \oplus G_2$,
5. if $xy \in E(G)$, then $G \setminus \{xy\} = G_1 \oplus G_2$.

Proof. Let $C_1, \dots, C_r$ be the components of $G - \{x, y\}$. Then $r \geq 2$. For each i , let H_i be the graph with vertex set $V(C_i) \cup \{x, y\}$ and edge set $\{wz \in E(G) : w, z \in V(H_i)\} \cup \{xy\}$. By lemma, each H_i is 2-connected, and $G_2 = H_2 \cup \dots \cup H_r$ is 2-connected by lemma (as $|V(H_i) \cap V(H_j)| \geq 2$). Set $G_1 = H_1$. Then indeed all of the five conditions hold. $\square$

Corollary 1.35. Let G be a 2-connected graph with ≥ 4 vertices. Let $e \in E(G)$. Then either $G \setminus e$ is 2-connected or G/e is 2-connected (or both).

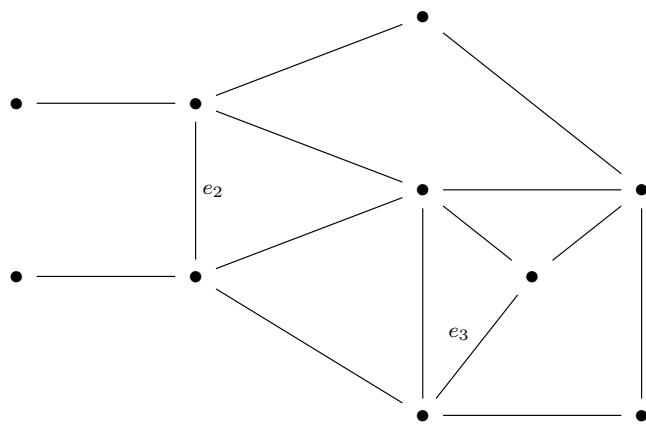
Proof. Fix $e = xy$. Suppose G/e is not 2-connected. Let z be the image of e under contraction. Since $|V(G/e)| \geq 3$, and G/e is connected, G/e has a cutvertex, say w . If $w \neq z$, then w is a cutvertex of G , contradiction. Hence $w = z$, and so $\{x, y\}$ is a vertex cut of G . Let G_1 and G_2 be as in the Decomposition theorem, then G_1, G_2 are 2-connected, and $G \setminus e = G_1 \oplus G_2$. Hence $G \setminus e$ is 2-connected since the 2-sum of 2-connected graphs is 2-connected. $\square$

Example 1.36. Suppose we have G and e_1, e_2, e_3 to be



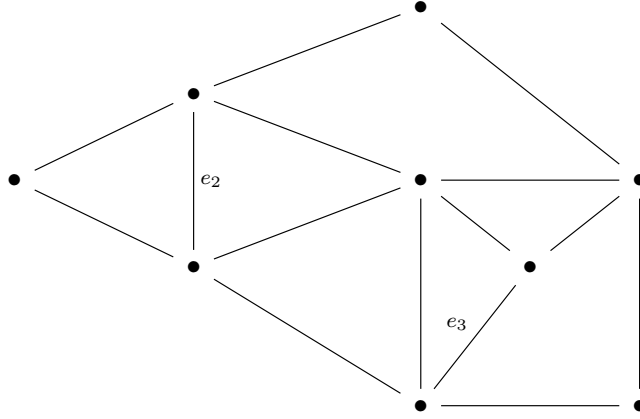
G

Then $G \setminus e_1$ is not 2-connected:



G

But G/e_1 is 2-connected:



G

Similarly, $G \setminus e_2$ is 2-connected, G/e_2 is not, and both $G \setminus e_3$ and G/e_3 are 2-connected.

Reducing Kuratowski's theorem to 3-connected graphs:

Theorem 1.37 (K2). Suppose that every 3-connected graph that does not contain a subdivision of K_5 or $K_{3,3}$ is planar. Then every 2-connected graph that does not contain a subdivision of K_5 or $K_{3,3}$ is planar.

Proof. We use induction on the number of vertices.

Base case: $|V(G)| \leq 5$. All graphs on ≤ 5 vertices except K_5 are planar. Hence any G satisfying the assumption is planar.

IH: Assume $|V(G)| \geq 6$, and every 2-connected graph H with no subdivision of K_5 or $K_{3,3}$ and $|V(H)| < V(G)$ is planar. Consider G , a 2-connected graph with no subdivision of K_5 or $K_{3,3}$.

- If G is 3-connected, then we are done.
- If G is 2-connected but not 3-connected, then there exists a vertex cut $\{x, y\} \in G$. By our Decomposition theorem, there exists 2-connected graphs G_1, G_2 , each containing the edge xy , such that $G = G_1 \oplus G_2$ or $G \setminus e = G_1 \oplus G_2$. Then $|V(G_1)| < V(G)$ (since $|V(G_2)| \geq 3$ by 2-connectedness). By Lemma 1.30, in $G_1 \oplus G_2$, there exists a subdivision of G_1 in $G_1 \oplus G_2$. Thus G_1 contains no subdivision of K_5 or $K_{3,3}$. By IH, G_1 is planar. Similarly, G_2 is planar. Take a planar drawing of G_1 with edge xy on the outer face and a planar drawing of G_2 with edge xy on the outer face. Gluing these two planar drawings along the edge xy gives a planar drawing of G (if $xy \in E(G)$), or a planar drawing of $G + e$. Hence, G is planar by induction.

□

Let H be a graph and C be a cycle, and $uv \in E(H) \cap E(C)$, with $|V(C)| = k$, then $H \oplus C$ is the same graph as the graph we get by subdividing the edge uv in H . The number of new path vertices is $k - 2$.

Theorem 1.38 (“building 2-connected graphs”). Let G be a 2-connected graph. Then at least one of the following holds:

- (a) G is a cycle,
- (b) there exists $e \in E(G)$ such that $G \setminus e$ is 2-connected,
- (c) G is the 2-sum of a 2-connected graph H and a cycle C . That is, G is the result of subdividing an edge of a 2-connected graph G' at least once.

Proof. We call a graph “good” if it satisfies at least one of the three conditions. Notice that any good graph is 2-connected. Let G be a given 2-connected graph, and let H be a good subgraph of G with $|E(H)|$ as large as possible (such H exists since G contains a cycle). If $|E(H)| = |E(G)|$, then we are done. Suppose that $|E(G)| > |E(H)|$.

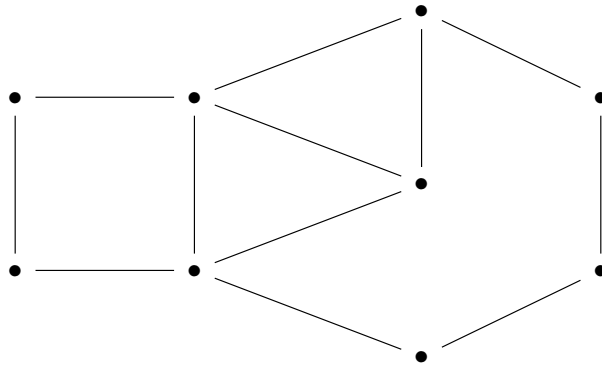
1. If $|V(H)| = |V(G)|$, then there exists $e \in E(G) \setminus E(H)$, and both endpoints of e are in $V(H)$. Then $G \setminus e$ contains H . So $G \setminus e$ is 2-connected (since $G \setminus e$ has same vertex set as H , and contains H). Hence G is good. Contradicts to the maximality of $|E(H)|$.
2. If $|V(H)| < |V(G)|$. Since G is connected, there exists $u \in V(H)$ and $x \in V(G) \setminus V(H)$, where $ux \in E(G)$ (as G is connected). Since u is not a cutvertex of G , there exists a path P in $G - u$ from x to some vertex $v \in V(H)$ such that $V(P) \cap V(H) = \{v\}$ (that is, P ends at the first vertex v it reaches in H). Let

$$H' = \begin{cases} H & , \text{ if } uv \in E(H) \\ H \cup \{uv\} & , \text{ if } uv \notin E(H) \end{cases}$$

Then H' is 2-connected by construction. Let C be the cycle $P \cup \{u\} \cup \{uv\}$. Then $J = H' \oplus C$ is a subgraph of G , is good (by (c)), and $|E(J)| = |E(H')| + |E(C)| - 2 \geq |E(H)| + 1$, contradicts to the maximality of $|E(H)|$.

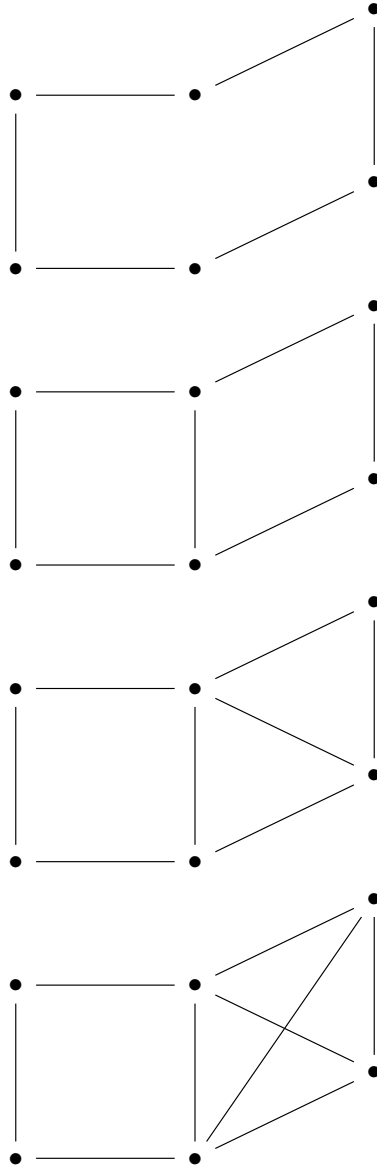
□

Example 1.39. We can “build” the 2-connected G



G

by the following steps:



Lemma 1.40. Let G be a 2-connected planar graph. In every planar drawing, every face is bounded by a cycle, and every edge is incident to exactly two distinct faces.

Proof. We will prove it by induction on $|E(G)|$. Since G has minimum degree ≥ 2 , then $|E(G)| = \frac{1}{2} \sum_{v \in V(G)} d(v) \geq |V(G)|$.

Base case: $|E(G)| = |V(G)|$. Since G is 2-connected, G is a cycle, and the result follows directly.

IH: Assume $|E(G)| > |V(G)|$ and for every 2-connected planar graph H with $|E(H)| < |E(G)|$ in every planar drawing of H , every face is bounded by a cycle, and every edge is incident to 2 faces. Fix a planar drawing $\tilde{G}$ of G . By our previous theorem, we split it into two cases:

Case 1: there exists $e = xy \in E(G)$ such that $H = G \setminus e$ is 2-connected. Then H has a planar drawing $\tilde{H}$, obtained by erasing e from the planar drawing of G . Then, there is a face F of $\tilde{H}$ that has both x and y on its boundary. By IH, every face of $\tilde{H}$ is bounded by a cycle, and every edge is incident to 2 faces. Then replacing e into $\tilde{H}$ to get $\tilde{G}$ partitions F (bounded by a cycle by IH) into 2 faces, each bounded by a cycle. So the drawing of G is as claimed

Case 2: G is obtained by subdividing an edge of a 2-connected graph H : then a planar drawing of H can be obtained from $\tilde{G}$ via the subdivision, then $\tilde{H}$ is “nice” by IH. So $\tilde{G}$ is also nice. $\square$

Lemma 1.41. Let G be a k -connected graph. Suppose G has a vertex cut W of size k . Then for every component C of $G - W$, and every $w \in W$, there is an edge in G from w to C .

Proof. Suppose, for contradiction, that some $w_0 \in W$ has no edge to some component C . Then, $W \setminus \{w_0\}$ is a vertex cut of G with size $k - 1$, contradicts to G is k -connected. $\square$

Definition 1.42. Let G be a 3-connected graph. An edge e of G is contractible if G/e is also 3-connected.

Theorem 1.43. Every 3-connected graph with at least 5 vertices has a contractible edge.

Proof. Suppose, for contradiction, that G/e is not 3-connected for any edge e . Since $|V(G)| \geq 5$, we have $|V(G/e)| \geq 4$. Also, G/e would be connected. Therefore, every G/e must have a vertex cut of size ≤ 2 . Let $e = xy$ be fixed and z be the image of e in G/e . Then, every vertex cut of size ≤ 2 of G/e must contain z , otherwise it would be a vertex cut of G of size ≤ 2 . Therefore, any vertex cut of G/e of size ≤ 2 has size 2, otherwise z is the cutvertex in G/e and hence $\{x, y\}$ is a vertex cut in G . Thus, for every edge xy in G , there exists $w \in V(G) \setminus \{x, y\}$ such that $\{z, w\}$ is a vertex cut of G/e . This implies $\{x, y, w\}$ is a vertex cut of G . We choose:

- the edge xy of G ,
- the vertex $w \in V(G) \setminus \{x, y\}$,
- the component C of $G - \{x, y, w\}$

such that $|V(C)|$ is as small as possible. By the previous lemma, we know that w has a neighbour v in C . Hence, there exists a vertex $u \in V(G) \setminus \{w, v\}$ such that $\{w, v, u\}$ is a vertex cut of G .

Claim: Some component D of $G - \{v, w, u\}$ is disjoint from $\{x, y\}$.

Proof. $G - \{v, w, u\}$ has at least 2 components. If there are at least 3 components, then we are done. If there are two components, since x, y are joined by an edge, they are in the same component (or u is one of x, y , and two components means there is a component without $\{x, y\} \setminus \{u\}$). $\square$

We will now show (w, v, u, D) contradicts our choice of (x, y, w, C) . By our previous lemma, we know v has a neighbour q in D . Then $q \in V(C)$ since the neighbours of v are $V(C) \cup \{x, y, w\}$, D does not contain x, y by choice, and D does not contain w . So $D \cap C \neq \emptyset$, hence $D \subset C$ since D is disjoint from $\{x, y, w\}$. Since $v \in V(C) \setminus V(D)$, we have $|V(D)| < |V(C)|$, contradicts to the minimality of $|V(C)|$. $\square$

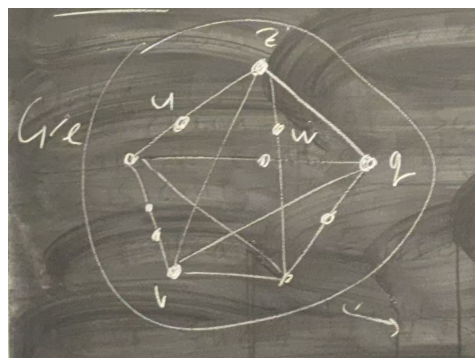
Lemma 1.44. Let e be an edge of a graph G , and suppose G/e contains a subdivision of K_5 . Then G contains a subdivision of K_5 or a subdivision of $K_{3,3}$.

Proof. Let z be the image of the edge $e = xy$ under contraction. Let K be a subdivision of K_5 in G/e . If $z \notin V(K)$, then K is a subgraph of G , so we are done.

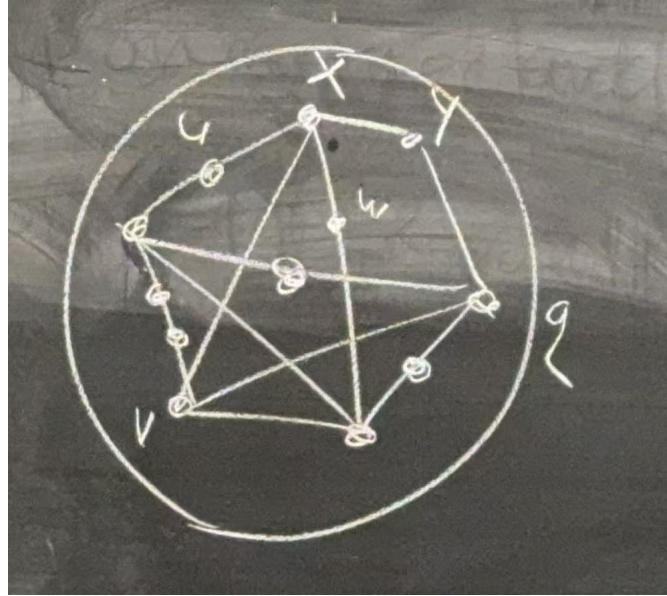
Case 1: z is a path vertex in K . Say the neighbours of z are u, v in K . Then in G , we have at least one of $x, y \sim u$ and at least one of $x, y \sim v$. Therefore, we get a subdivision of K_5 in any cases.



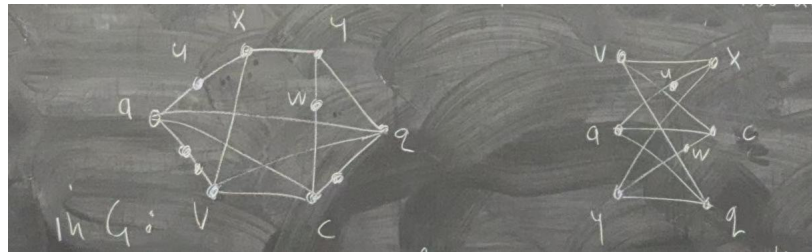
Case 2: z is a branch vertex in K . Let u, v, w, q be the neighbours of z in K . Each of $\{u, v, w, q\}$ was a neighbour of at least one of $\{x, y\}$ in G .



If at least 3 of $\{u, v, w, q\}$ was joined to, say x , then we find a subdivision of K_5 in G



If two of $\{u, v, w, q\}$ are joined to x and two are joined by y ,



We find a subdivision of $K_{3,3}$ in G .

□

Theorem 1.45 (Kuratowski's Theorem for 3-connected graphs; K3). Let G be a 3-connected graph that does not contain a subdivision of K_5 or $K_{3,3}$. Then G is planar.

Proof. We will prove it by induction on $|V(G)|$.

Base Case: $|V(G)| \leq 5$. Then, since K_5 is the only non-planar graph with at most 5 vertices, we are done.

IH: Assume $|V(G)| \geq 6$, and every 3-connected graph with fewer than $|V(G)|$ vertices that does not contain a subdivision of K_5 or $K_{3,3}$ is planar.

Since $|V(G)| \geq 6$, we know, by our previous theorem, that G has a contractible edge e . So $H = G/e$ is 3-connected and has at least 5 vertices, and $|V(H)| < |V(G)|$.

Claim: $H = G/e$ does not contain a subdivision of K_5 or $K_{3,3}$.

Proof. We split it into two cases:

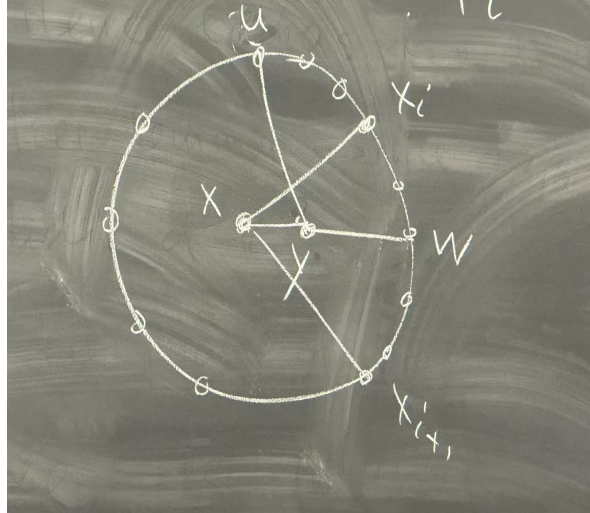
1. If G/e contains a subdivision of K_5 , then by our earlier lemma, G contains a subdivision of K_5 or $K_{3,3}$. Contradicts.
2. If G/e contains a subdivision of $K_{3,3}$, then in particular, G/e has $K_{3,3}$ as a minor. Hence G has $K_{3,3}$ as a minor. Hence, by Lemma 1.26, since $K_{3,3}$ is cubic, we find G contains a subdivision of $K_{3,3}$. Contradicts.

□

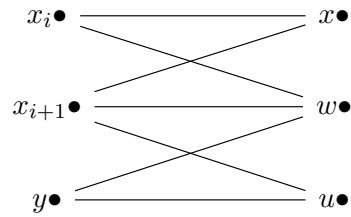
Hence by IH, we know $H = G/e$ is planar. Let $\tilde{H}$ be a planar drawing of $H = G/e$. Let $e = xy$ and let z denote the image of e under contraction. Erase z (and its incident edges) from $\tilde{H}$ to get a planar drawing of $H - z = G - \{x, y\}$. Since H is 3-connected, $H - z$ is 2-connected. Then by an earlier lemma, we know that in $\tilde{H} - z$, every face is bounded by a cycle. So the face of $\tilde{H} - z$ that contained z in $\tilde{H}$ is bounded by a cycle C . Since all neighbours of x and of y in G became neighbours of z in H , we know that all such neighbours are vertices of C . Let $x_1, \dots, x_t$ denote the neighbours of x (except y) in order as they appear in C . Let P_i denote the path from x_i to x_{i+1} on C , for $1 \leq i \leq t$ ($t+1 := 1$). We consider where on C the neighbours of y lie.

Case 1: All neighbours $N(y) \setminus \{x\}$ lie in the same P_i . Then we can complete the drawings to a planar drawing of G .

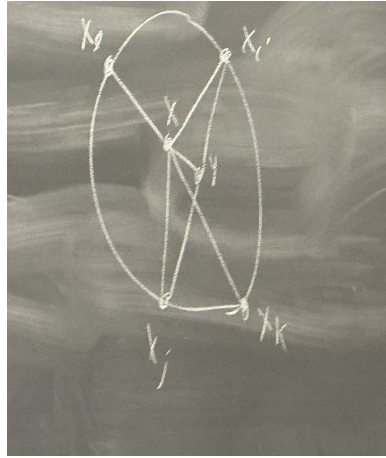
Case 2: y has a neighbour in the interior of some P_i and a neighbour not in P_i . This contains a subdivision of $K_{3,3}$:



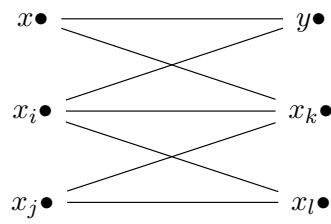
This contains a subdivision of $K_{3,3}$:



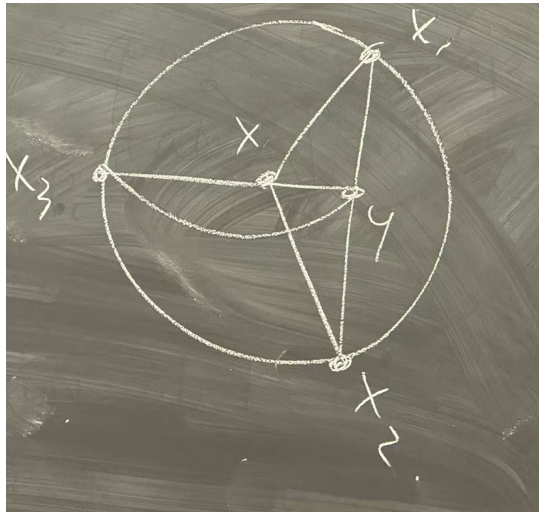
Case 3 : y has neighbours x_i and x_j where x_i, x_j are not both endpoints of the same P_l :



This contains a subdivision of $K_{3,3}$ as well:



Case 4: y has 3 neighbours x_1, x_2, x_3



This contains a subdivision of K_5 .

Therefore, only Case 1 can happen, therefore G is planar. $\square$

Theorem 1.46 (Kuratowski's Theorem for general graphs). A graph is planar if and only if it does not contain a subdivision of K_5 or $K_{3,3}$.

Theorem 1.47. ($\implies$) From Math 239, K_5 and $K_{3,3}$ are both non-planar, so any graph with a subdivision of K_5 or $K_{3,3}$ as a subgraph is non-planar.

($\impliedby$) Suppose G does not contain a subdivision of K_5 or $K_{3,3}$.

1. If G is 3-connected, it is planar by the previous theorem.
2. If G is 2-connected, it is planar by Theorem K2.
3. If G is connected by not 2-connected. Since every block of G is a single vertex, K_2 , or 2-connected, and the 2-connected blocks are planar by K_2 , all blocks of G are planar, by Theorem K1, G is planar.
4. If G is not connected, consider the components.

1.4 General Connectivity

Definition 1.48. A set of paths $\{P_1, \dots, P_k\}$ in a graph G , all with the same end points x and y , is said to be internally disjoint if $V(P_i) \cap V(P_j) = \{x, y\}, \forall i \neq j$.

Definition 1.49. Let $x, y \in V(G)$, we say a vertex cut W of G is said to separate x, y in G if x, y in different components of $G - W$.

Remark 1.50. $x, y \notin W, xy \notin E(G)$.

Remark 1.51. If there exists a set of k ID (internally disjoint) (x, y) -paths in G , then there $\nexists$ a vertex cut W with $|W| \leq k - 1$ that separates x, y in G .

Theorem 1.52 (Menger's Theorem). Let a, b be non-adjacent vertices in G . Let s be a minimal size of a vertex cut separating a, b in G . Then G contains a set of s ID (a, b) -path.

Proof. If $s \leq 1$, then it is clearly true. Assume $s \geq 2$. We prove it by induction on $|E(G)|$.

Base case: $|E(G)| = 0$, vacuously true.

IH: Assume $|E(G)| \geq 1$, every graph H with $|E(H)| < |E(G)|$, every pair of non-adjacent vertices $x, y \in V(H)$, there exists t ID (x, y) -path where $t = \text{minimal size of a vertex cut separating } x, y \text{ in } H$.

Given $G, a, b \in V(G)$ with $ab \notin E(G)$.

Case 1: Every edge of G is incident to a or b . Let

$$N(a) = \{x \in V(G) : ax \in E(G)\}$$

denoted the neighbours of a in G . Then $s = |N(a) \cap N(b)|$. So $\{aub : u \in N(a) \cap N(b)\}$ is a set of s ID (a, b) -paths.

Case 2: There exists an edge $e = xy$ in G with $\{x, y\} \cap \{a, b\} = \emptyset$. Let $H = G \setminus e$. Let S be a vertex cut of H of minimal size that separates a, b in H . If $|S| = s$, then by IH, H contains a set of s ID (a, b) -paths, hence in G . We may assume that $|S| < s$. Note that $S \cup \{x\}$ is a vertex cut separating a, b in G , then $|S \cup \{x\}| \geq s$, hence $|S \cup \{x\}| = s$, $|S| = s - 1$. We know there exists an (a, b) -path P in $G - S$ since $|S| < s$, so it must contain xy . Assume WLOG x is closer to a on the path than y . Let G_b be the graph obtained from G by contracting all edges in the component C_b of $G - (S \cup \{x\})$ that contains b . Let z_b be the image under contraction, then $|E(G_b)| < |E(G)|$ since the edges of P from x to b is disappeared (shrunk to 1). If T is a vertex cut separating a and z_b in G_b , then T is a vertex cut separating a and b in G . Then $|T| \geq s$, that is, the minimal size of vertex cut separating a, z_b in G_b is at least s . Hence by IH, there exists a set of $\geq s$ ID (a, z_b) -paths in G_b . Since $N(z_b)_{G_b} = S \cup \{x\}$, exactly s of these paths $P_1, \dots, P_s$. WLOG, P_1 contains x , P_i contains $w_i \in S$ for $i = 2, \dots, s$. Similarly, we can find s ID (z_a, b) -paths in the graph G_a , call them $Q_1, \dots, Q_s$.

Claim: $P_1 \cup Q_1 \cup \{xy\}, P_2 \cup Q_2, \dots, P_s \cup Q_s$ form a set of s ID (a, b) -paths in G .

Proof. Since each $V(P_i) \setminus (S \cup \{x, z_b\})$ is contained in component C_a of $H - S$, and similarly for C_b, Q_j . We know they are ID. □

□

Theorem 1.53 (Menger-Whitwy Theorem). Let G be a graph with $|V(G)| \geq 2$. Then G is k -connected iff $\forall$ pair of vertices a, b , $\exists$ a set of k ID (a, b) -path.

Proof. It is clearly true for $k = 1$. We may assume $k \geq 2$.

($\implies$) Assume G is k -connected. Let $a, b \in V(G)$. If $ab \notin E(G)$, then by Menger's theorem, there exists a set of k ID (a, b) -paths. If $e = ab \in E(G)$, then we claim that $G' = G \setminus e$ is $(k - 1)$ connected.

Proof. Suppose not, then there exists a vertex cut Y of $G \setminus e$ with $|Y| \leq k - 2$, then a, b are in different components of $G' - Y$, otherwise Y is a vertex cut separating a, b . WLOG, some components of $G' - Y$ not containing b has a vertex $c \neq a$ (since $|V(G)| = |V(G \setminus e)| \geq k + 1$). Then $Y \cup \{a\}$ is a vertex cut of G separates b, c , contradicts to G is k -connected. □

By Menger's Theorem, G' contains a set of $k - 1$ ID (a, b) -paths. Together with $e = ab$, we obtain k ID (a, b) -paths.

($\impliedby$) : Assume for G : $\forall$ pair vertices is joined by a set of k ID paths, then G has no vertex cut of size $\leq k - 1$.

To see $|V(G)| \geq k + 1$: let $a, b \in V(G)$ of the k ID (a, b) -paths, at least $k - 1$ of them have an internal vertex, all of which are distinct. Therefore

$$|V(G)| \geq 2 + k - 1 = k + 1$$

Hence, G is k -connected. □

Lemma 1.54 (Extension Lemma). Let G be a k -connected graph. Let $Y \subset V(G)$ be a set of vertices with $|Y| = k$, then the graph H with $V(H) = V(G) \cup \{x\}$, $E(H) = E(G) \cup \{xy : y \in Y\}$ is k -connected, where x is a new vertex.

Proof. Firstly, $|V(H)| > |V(G)| \geq k + 1$. Suppose, for contradiction, that there exists a vertex cut W of H with $|W| \leq k - 1$. If $x \in W$, then $W \setminus \{x\}$ is a vertex cut of G of size $\leq k - 2$, contradicting k -connectivity of G . So $x \notin W$. Let $z \in V(G)$ be a vertex in a component of $H - W$ that does not contain x . Let y be a vertex in $Y \setminus W$ (which exists because $|Y| > |W|$). Then y is in the component of x in $G - W$, hence W is a vertex cut of G separating y and z , contradicts to the k -connectivity of G . $\square$

Definition 1.55. Let G be a graph, $x \in V(G)$ and $Y \subseteq V(G) \setminus \{x\}$. An (x, Y) -fan in G is a set

$$S = \{P_1, \dots, P_k\}$$

of paths in G from x to Y , such that $k = |Y|$, and $V(P_i) \cap V(P_j) = \{x\}$ for each $i \neq j$.

Lemma 1.56 (Fan Lemma). Let G be a k -connected graph. Let $x \in V(G)$, $Y \subseteq V(G) \setminus \{x\}$, where $|Y| = k$. Then G contains an (x, Y) -fan.

Proof. Let H be the graph with $V(H) = V(G) \cup \{z\}$, and $E(H) = E(G) \cup \{zy : y \in Y\}$. By our extension lemma, H is k -connected. By the Menger-Whitney Theorem, there exists a set of k ID (x, z) paths $\{P_1, \dots, P_k\}$ in H . Since $N(z) = Y$, each path P_i must end with an edge yz , where $y \in Y$, and these are all distinct. Hence $\{Q_1, \dots, Q_k\}$ is an (x, Y) -fan where $Q_i = P_i - \{z\}$. $\square$

Remark 1.57. Fan Lemma is indeed a “if and only if” lemma.

Lemma 1.58 (Cycle Lemma). Let G be a k -connected graph with $k \geq 2$. Let $Y \subset V(G)$ be such that $|Y| = k$, then G contains a cycle C with $Y \subset V(C)$.

Proof. Let C_0 be a cycle contains as many vertices of Y as possible, say $V(C_0) \supseteq \{y_1, \dots, y_m\}$. Then $m \geq 2$ by our characterization of 2-connectivity. If $m = k$, we are done. So suppose, for contradiction, that $m < k$.

Case 1: $V(C_0) = \{y_1, \dots, y_m\}$. Let $y \in Y \setminus V(C_0)$. By the Fan Lemma for m -connected graphs (we know G is k -connected, therefore m -connected), we can find a $(y, \{y_1, \dots, y_m\})$ -fan in G . Say the path from y to y_1 is P_1 , to y_2 is P_2 . Then $C_0 \setminus (y_1 y_2) \cup P_1 \cup P_2$ is a new cycle in G containing $\geq m + 1$ vertices in Y . Contradicts to the choice of C_0 .

Case 2: $V(C_0) \setminus \{y_1, \dots, y_m\} \neq \emptyset$. Choose $x \in V(C_0) \setminus \{y_1, \dots, y_m\}$. Let $\{P_0, \dots, P_m\}$ be a $(y, \{y_1, \dots, y_m, x\})$ -fan in G (note $m + 1 \leq k$). For $0 \leq i \leq m$, let u_i be the first vertex of P_i on C_0 when going from y to C_0 (possibly $u_i = y_i$). So we get $m + 1$ u_i 's and m y_i 's on $V(C_0)$. By the Pigeonhole Principle, some segment y_i to y_{i+1} on C_0 contains two vertices u_j and u_l . Then replacing the (u_j, u_l) -segment with the (y_i, y_{i+1}) -segment of C_0 with the (u_j, y) -segment of P_j , and the (u_l, y) -segment of P_l gives a cycle containing $y_1, \dots, y_m, y$. Contradicts to the choice of C_0 . $\square$

Remark 1.59. The converse of the Cycle Lemma is not generally true. Consider the 100 cycle C_{100} , then every subset of $V(C_{100})$ with 98 vertices is contained in a cycle $C = C_{100}$, but C_{100} is not 98-connected.

Theorem 1.60. Every 3-connected graph contains a subdivision of K_4 .

Proof. Fix $x \in V(X)$. Then $G - x$ is 2-connected. By the Cycle Lemma, $G - x$ contains a cycle C . Let $y_1, y_2, y_3 \in V(C)$ be distinct vertices (or, by the previous results, $G - x$ contains a cycle). By the Fan Lemma, there exists (x, Y) -fan, where $Y = \{y_1, y_2, y_3\}$. Say P_1, P_2, P_3 are the paths from x to y_1, y_2, y_3 . Let u_i be the first vertex in P_i that is also in C (u_i 's exist because we can take u_i to be y_i , and are distinct because P_i are disjoint). Let Q_i be the segment of P_i from x to u_i . Then $C \cup Q_1 \cup Q_2 \cup Q_3$ is a subdivision of K_4 in G . $\square$

2 Matchings

Definition 2.1. A matching in a graph G is a set M of disjoint edges in G . A maximal matching in G is a matching of maximum size in G . We denote the size of a maximum matching in G is written $\nu(G)$. The matching M saturates the vertex $v \in V(G)$ if v is incident to an edge of M , otherwise we say v is exposed (or unsaturated). We say M is a perfect matching if M saturates $V(G)$.

Definition 2.2. A vertex cover W of a graph G is a set $W \subseteq V(G)$ such that $G - W$ has no edges. We write $\tau(G)$ for the minimal size of a vertex cover of G .

Remark 2.3. In any graph, $\tau(G) \geq \nu(G)$.

2.1 The Greedy Algorithm

INPUT: Graph G

OUTPUT: Matching M in G .

1. Set $M := \emptyset, H := G$
2. IF H has no edges, STOP, OUTPUT M
3. ELSE choose $xy \in E(H)$, set $M := M \cup \{xy\}$, $H := H - \{x, y\}$, go to 2.

It is easy to compute, but it could be far from the maximal matchings.

Lemma 2.4. The Greedy Algorithm always finds a matching in G of size $\geq \frac{\nu(G)}{2}$

Proof. Let M be a matching obtained by the Greedy Algorithm. Then the set $V(M)$ of vertices saturated by M is a vertex cover of G . Let M^* be a maximal matching in G . Since each edge of M^* has a vertex in $V(M)$, and no two edges of M^* share the same vertex, we find $|M^*| \leq |V(M)| = 2|M|$, so

$$|M| \geq \frac{|M^*|}{2} = \frac{\nu(G)}{2}.$$

$\square$

2.2 Bipartite Matching

Let G be a bipartite graph with vertex classes X and Y . Notice that if $S \subseteq X$ is such that $|N(S)| < |S|$, then there is no matching saturating X .

Theorem 2.5 (Hall's Theorem). Let G be a bipartite graph with vertex classes X and Y . Then G has a matching saturating X if and only if

$$|N(S)| \geq |S|, \forall S \subseteq X \quad (*) \text{ ("Hall's condition")}$$

Proof. ($\implies$): as above

($\impliedby$): Assume G has $(*)$, we use induction on $|X|$:

Base case: $|X| = 1$, then G has a matching of size 1, and saturating X .

IH: Assume $|X| \geq 2$, and for every bipartite graph H with vertex classes X' and Y' , where $|X'| < |X|$, that satisfies $(*)$ for H, X' , there exists a matching saturating X' .

Given G ,

Case 1 Every $S \subseteq X$ with $\emptyset \neq S \neq X$ satisfies $|N(S)| > |S|$.

Let $xy \in E(G)$. Let $H := G - \{x, y\}$. For every $\emptyset \neq S \subseteq X' = X \setminus \{x\}$, we have $N_H(S) = N_G(S) \setminus \{y\}$. So

$$|N_H(S)| \geq |N_G(S)| - 1 > |S| - 1 \text{ (by assumption of Case 1)}$$

So $|N_H(S)| \geq |S|$. This says Hall's condition holds in H , so by IH, H has a matching M' saturating X' , then $M' \cup \{xy\}$ saturated X .

Case 2 There exists S_1 with $\emptyset \neq S_1 \neq X$ such that $|S_1| = |N(S_1)|$. Let G_1 be the subgraph of G induced by S_1 and its neighbourhood $N(S_1)$. Then for each $S \subset S_1$, by the assumption on G , we have

$$|N_{G_1}(S)| = |N_G(S)| \geq |S|$$

So by IH ($|S_1| < |X|$), there exists a matching M_1 in G_1 saturating S_1 .

Let G_2 be the subgraph of G induced by $(X \setminus S_1) \cup (Y \setminus N(S_1))$. Take an arbitrary $S \subseteq X \setminus S_1$, then

$$N_{G_2}(S) = N_G(S \cup S_1) \setminus N_G(S_1)$$

So $|N_{G_2}(S)| = |N_G(S \cup S_1)| - |N_G(S_1)|$, but $|N_G(S \cup S_1)| \geq |S \cup S_1|$ by assumption $(*)$ on G , and $|S_1| = |N(S_1)|$. So we get

$$|N_{G_2}(S)| \geq |S \cup S_1| - |S_1| = |S|.$$

So $(*)$ holds for G_2 , by IH, we get a matching M_2 in G_2 saturating $X \setminus S_1$, then $M_1 \cup M_2$ saturates X .

□

Theorem 2.6. Any regular bipartite graph G of positive degree k has a perfect matching.

Proof. Let X and Y be the vertex classes of G . Let $S \subseteq X$ be arbitrary. Let $E(S, N(S))$ denotes the set of edges of G joining S to $N(S)$. Then,

$$|E(S, N(S))| = k|S|$$

since G is k -regular. Similarly, $|E(N(S), X)| = k|N(S)|$, and

$$|E(S, N(S))| \leq |E(X, N(S))| = k|N(S)|$$

Therefore $k|S| = |E(S, N(S))| \leq k|N(S)|$. Therefore, since $k > 0$, $|N(S)| \geq |S|$. Hence by Hall's Theorem, G has a matching saturating X . Since $E(G) = k|X| = k|Y|$, we have $|X| = |Y|$, therefore this is a perfect matching. $\square$

Theorem 2.7 (defect version of Hall's Theorem). Let $d \geq 0$ and let G be a bipartite graph with vertex classes X and Y , then G has a matching of size $\geq |X| - d$ if and only if

$$|N(S)| \geq |S| - d \text{ for each } S \subseteq X \quad (**)$$

Proof. ($\implies$): If G has a matching M of size $\geq |X| - d$, then for every $S \subseteq X$, M saturates at least $|S| - d$ vertices of S to vertices in $N(S)$.

($\impliedby$): Assume G has (**). Form the graph H by adding a set D with d vertices to Y and joining all vertices in D to all vertices in X . Let $\emptyset \neq S \subseteq X$, then

$$N_H(S) = N_G(S) \cup D$$

$$\implies |N_H(S)| = |N_G(S)| + d \geq |S| - d + d = |S|.$$

Hence by Hall's Theorem, H has a matching M saturating X . At most d vertices of M have a vertex in D , so the rest of M forms a matching of size $\geq |S| - d$ in G . $\square$

Theorem 2.8 (König's Theorem). If G is bipartite, then $\tau(G) = \nu(G)$.

Proof. Let X, Y be the vertex classes of G . Let x, y be new vertices, and form H via

$$\begin{aligned} V(H) &= V(G) \cup \{x, y\} \\ E(H) &= E(G) \cup \{xv : v \in X\} \cup \{yw : w \in Y\} \end{aligned}$$

Let W be a vertex cut separating x and y in H . Then $W \subset V(G)$ since $x, y \notin W$, and W is a vertex cover of G (any edge survives would give a path from x to y). So $|W| \geq \tau(G)$. Then by Menger's Theorem, there is a set of at least $\tau(G)$ ID (x, y) -paths in H . So taking the second edge on each of these paths gives a matching in G of size $\geq \tau(G)$. Hence $\nu(G) \geq \tau(G)$, hence $\nu(G) = \tau(G)$. $\square$

Definition 2.9. A set $U \subseteq V(G)$ in G is called independent in G if there is no edge in G joining two vertices of U . The maximum size of an independent set in G is denoted $\alpha(G)$.

Definition 2.10. A set $F \subseteq E(G)$ is called an edge cover of G if every vertex of G is incident to an edge of F . We write $\rho(G)$ for the minimum size of an edge cover of G .

Lemma 2.11. For every graph G , $\alpha(G) + \tau(G) = |V(G)|$.

Proof. For any vertex cover W of G , the set $V(G) \setminus W$ is independent, and vice versa. Hence W is minimal if and only if $V(G) \setminus W$ is a maximum independent set. $\square$

Lemma 2.12 (Gallai Lemma). For every graph G with no isolated vertices,

$$\rho(G) + \nu(G) = |V(G)|$$

Proof. Set $|V(G)| = n$. Let M be a matching with $|M| = \nu(G)$. Then $V(G) \setminus V(M)$ is independent in G (otherwise there is a bigger matching). Construct an edge cover of G as follows:

for each $x \notin V(M)$, take an edge incident to x , add M to this set.

This gives $\rho(G) \leq |M| + (n - |V(M)|) = |M| + n - 2|M| = n - |M| = n - \nu(G)$. Hence

$$\rho(G) + \nu(G) \leq n.$$

We now want to show that $\rho(G) + \nu(G) \geq n$. Let F be a minimum edge cover of G . Let $H := (V(G), F)$ be a subgraph of G . Then, each edge of H is incident to a vertex of degree 1 in H . Thus, H has no cycles and is thus a forest. Recall that, for any tree T , $E(T) = V(T) - 1$ and, for any forest T , $E(T) = V(T) - c$, where c is the number of components of T . Thus, H has $V(H) - E(H) = n - \rho(G)$ components, of which each has ≥ 2 vertices and ≥ 1 edges since F is an edge cover.

Thus, by taking one edge from each component of H , we get a matching of size $n - \rho(G)$. Hence

$$\nu(G) \geq n - \rho(G).$$

$\square$

Definition 2.13. Let M be a matching of G . An M -alternating path of G is a path in G of which every second edge is in M . Note that it does not matter if M -edges are the even- or odd-indexed edges.

Definition 2.14. Let M be a matching of G . An M -augmenting path of G is an M -alternating path P in G with length ≥ 1 and begins and ends with an M -exposed vertex. Notice that $|E(P)|$ is always odd.

Lemma 2.15 (Switching). Let M be a matching of G . Let G contains an M -augmenting path P . Then

$$M' := M \triangle E(P) := (M \setminus E(P)) \cup (E(P) \setminus M)$$

is a matching in G larger than M . This operation is called switching on P .

Lemma 2.16 (Properties of Two Matchings). Let M_1, M_2 be matchings in G . Define the subgraph $H := (V(G), M_1 \cup M_2)$. Then we have that

- H has a maximal degree that is ≤ 2 .
- Every component of H is either a path or a cycle.

- Every cycle of H is even.
- Every path of H alternates between M_1 and M_2 .
- In each path, the number of M_1 -edges and M_2 -edges differ by ≤ 1 , or it is a single edge in $M_1 \cap M_2$.

Theorem 2.17 (Berge). Let M be a matching in G . Then M is maximal $\iff$ there does not exist an M -augmenting path in G .

Proof. ($\implies$) Done. ($\impliedby$) Let M^* be a maximum matching in G . If $|M^*| = |M|$, we are done. So assume $|M^*| > |M|$. Let $H := (V(G), M^* \cup M)$. Then there is a component of H that is an odd path starting with M^* edge. Since every path in H must be alternating, it ends in M^* , therefore it is M -augmenting path, contradicts to the assumption. $\square$

Theorem 2.18 (Erdős-Pósa). For all graphs G , we have that $\nu(G) \geq \min\{\delta(G), \lfloor \frac{|V(G)|}{2} \rfloor\}$.

Proof. Let M be a maximum matching in G . If $|M| = \lfloor \frac{|V(G)|}{2} \rfloor$, we are done. So suppose otherwise. Then there are at least 2 M -exposed vertices in G . Let v and w be these vertices. Then, all neighbours of v and w are in the set $V(M)$ of M -saturated vertices due to the maximality of M . By Berge's Theorem, there cannot exist any M -augmenting paths in G . Let $xy \in M$. If $vy \in E(G)$, then $wx \notin E(G)$ otherwise $vyxw$ is M -augmenting. Similarly, if $vx \in E(G)$, we cannot have $wy \in E(G)$ for similar reasons. Thus, there are at most two edges that join the vertex set $\{v, w\}$ to the vertex set $\{x, y\}$. This holds for every edge of M , and so we have that $\deg(v) + \deg(w) \leq 2|M|$, thus

$$2\delta(G) \leq \deg(v) + \deg(w) \leq 2|M| \implies |M| \geq \delta(G).$$

$\square$

Definition 2.19. Let $G := ((X, Y), E(G))$ be bipartite. A set L of preference lists of G consists of a linear order $L(z)$ of $N(z)$ for each $V(G)$. We say that a vertex x prefers a vertex y' to another vertex y in G if

$$y' > y \text{ in } L(x)$$

Definition 2.20. A matching M is stable if for every edge $xy \notin M$, either

- x prefers its neighbour in M to y , or (i.e., $xy' \in M$ for some $y' > y$ in $L(x)$)
- y prefers its neighbour in M to x . (i.e., $x'y \in M$ for some $x' > x$ in $L(y)$)

2.3 The Gale-Sharpley Algorithm

INPUT: Bipartite graph G , vertex classes X and Y , preference lists L .

OUTPUT: A stable matching M^* in G .

1. Set $K(x) := L(X)$ for all $x \in X$. Set $M := \emptyset$.

2. IF for each $x \in X$, either
 - (i) $K(X) = \emptyset$, or
 - (ii) M saturates x ,
 then STOP.
3. ELSE choose $x \in X$ M -exposed and $K(x) \neq \emptyset$. For $y = \max(K(x))$
4. IF y prefers x to x' where $x'y \in M$: set $M := (M \setminus \{x'y\}) \cup \{xy\}$
5. ELSE if y is M -exposed, set $M := M \cup \{xy\}$
6. Set $K(x) := K(x) \setminus \{y\}$
7. Go to 2.

This algorithm terminates because

$$\sum_{x \in X} |K(x)| \text{ starts at value } \sum_{x \in X} l(x) \text{ and decreases at each step.}$$

correctness. Steps 4 and 5 ensures M^* is a matching ($\deg(x), \deg(y) \leq 1$). Notice that at each iteration, the situation

- improves or stays the same for every $y \in Y$,
- deteriorates or stays the same for every $x \in X$.

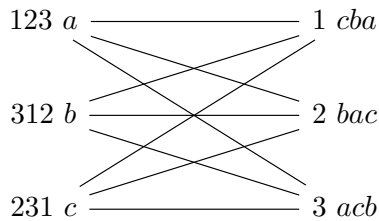
Consider an edge $x_0y_0 \notin M^*$. Then $y_0 \in L(x_0)$ and hence in $K(x_0)$ at initialization.

Case 1 x_0 proposes to y_0 . Then y_0 is improved, that is, y_0 is matched in the final matching to some x' that y_0 prefers to x_0 .

Case 2 x_0 never propose to y_0 . Then the algorithm terminates before x_0 reached y_0 on its list, that is, it is matched in M^* to some y' it prefers to y_0 .

□

Remark 2.21. There can be many stable matching in (G, L) :



has stable matchings

- $\{a1, b3, c2\}$ is stable (GS with X)
- $\{c1, b2, a3\}$ is stable (GS with Y)
- $\{a2, b1, c3\}$ is stable

Theorem 2.22. Let G be a bipartite graph with vertex classes X and Y , preference list L . Then all stable matchings in (G, L) saturate the same set of vertices.

Proof. Let M_1, M_2 be distinct stable matchings in (G, L) . Let H be a subgraph of G with $V(H) = V(G)$ and $E(H) = M_1 \cup M_2$. If M_1 and M_2 do not saturate the same set of vertices, then some component of H is an alternating path, say $x_1y_1, x_2y_2, \dots$, where (WLOG) x_1 is M_2 -exposed, and $x_1y_1 \in M_1$. Then since $x_1y_1 \notin M_2$, M_2 stable, then y_1 prefers (WLOG) x_2 to x_1 , where $x_2y_1 \in M_2$. Similarly, $x_2y_1 \notin M_1$ therefore $x_2y_1 \notin M_1$, therefore x_2 prefers y_2 to y_1 . By doing it inductively, $x_iy_{i-1} \notin M_1 \implies x_i$ prefers y_i to y_{i-1} , $x_iy_i \notin M_2 \implies y_i$ prefers x_{i+1} to x_i . This shows that no x_i can be M -exposed and no y_i can be M_2 -exposed. So this path never ends, contradicts. Hence M_1 and M_2 saturates the same set of vertices. $\square$

Corollary 2.23. All stable matchings in (G, L) are of the same size.

Definition 2.24. Let G be a bipartite graph with vertex classes X and Y , and preference lists L . A stable matching M_0 in G is a X -optimal if for every stable matching M in (G, L) , and every $x \in X$,

- if $xy \in M$, then there exists y' such that $xy' \in M_0$ and $y' \geq y$ in $L(X)$. That is, either $y' = y$ or x prefers y' to y .

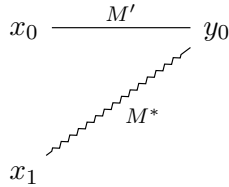
Similarly, M_0 is X -pessimal means for every $x \in X$,

- if $xy \in M$, then there exists y' with $xy' \in M_0$ and $y' \leq y$ in $L(X)$.

Note that X -optimal (X -pessal) matching is unique if it exists.

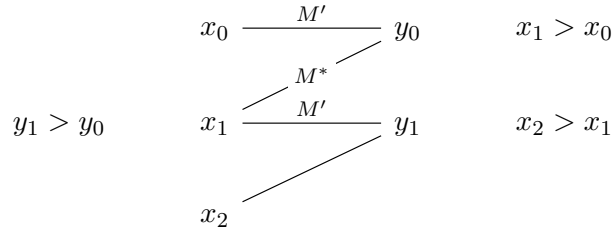
Proposition 2.25. The G-S stable matching with X making the proposals is X -optimal for every bipartite graph G , vertex classes X, Y , preferences L .

Proof. Suppose, for contradiction, that there exists some stable matching M' in G , and an edge $x_0y_0 \in M'$ (where $x_0y_0 \notin M^*$) such that x_0 prefers y_0 to its partner in M^* . Since $x_0y_0 \notin M^*$ and M^* stable, y_0 is matched in M^* to some x_1 where $x_1 > x_0$ in $L(y_0)$.



Let I be the earliest iteration in the GS algorithm in which we put an edge x_1y_0 into the matching M , where $x_0y_0 \in M'$ and x_0 strictly prefers y_0 to its partner in M^* . Since $x_1y_0 \notin M^*$ and y_0 prefers x_1 to x_0 , we know that x_1 is matched in M' to y_1 that is prefers to y_0 . In iteration I ,

x_1 proposes to y_0 and is accepted. Since x_1 prefers y_1 to y_0 , in some earlier iteration I_1 , x_1 was rejected by y_1 , since in some earlier iteration I_1 , y_1 received and accepted an offer from some x_2 , where $x_2 > x_1$ in $L(y_1)$. So in I_1 , $x_2 y_1$ was put into M , where $x_1 y_1 \in M'$, and x_1 strictly prefers y_1 to its partner in M^* , which contradicts our choice of I , so $x_0 y_0$ does not exist. Therefore M^* is X -optimal.

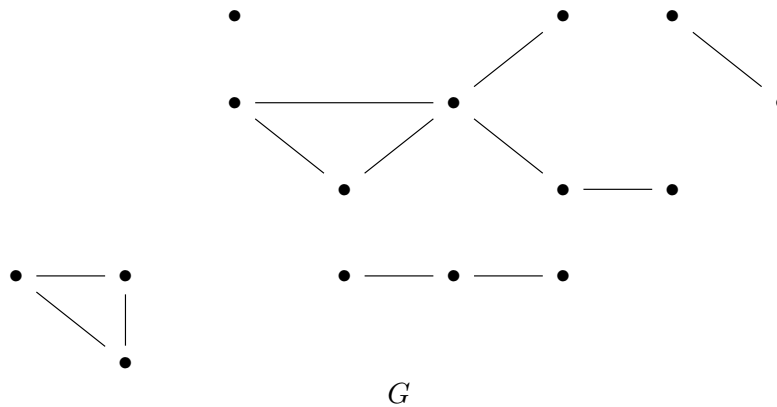


□

2.4 Perfect Matchings in General Grpahs

Definition 2.26. An odd component of a graph G is a component of G with an eodd number of vertices. The parameter $\text{odd}(G)$ is the number of odd components in G .

Example 2.27. For the following graph G , we have $\text{odd}(G) = 3$.



Proposition 2.28. Suppose a graph G has a perfect matching M . Then for any subset $T \subseteq V(G)$, there is an edge of M from each odd component of $G - T$ to T . Hence in particular, $|T| \geq \text{odd}(G - T)$. So if G has a perfect matching, then **for every $T \subseteq V(G)$, $|T| \geq \text{odd}(G - T)$** (called “Tutte’s Condition”). We’ll prove Tutte’s Theorem, which says this is if and only if.

Remark 2.29. If G has a perfect matching, then $|V(G)|$ is even, as Tutte’s Condition with $T = \emptyset$ implies $\text{odd}(G) = 0$, so every component of G is even, hence $|V(G)|$ is even.

Lemma 2.30. Let G be a graph that satisfies TC, that is,

$$\text{odd}(G - T) \leq |T|, \forall T \subseteq V(G).$$

If H is a graph with $V(H) = V(G)$ and $E(G) \subseteq E(H)$, then H satisfies TC, that is,

$$\text{odd}(H - T) \leq |T|, \forall T \subseteq V(H).$$

Proof. Let $T \subseteq V(H) = V(G)$ be arbitrary. Then adding edges inside T , or inside a component of $G - T$, or crossing from T to $G - T$ does not change $\text{odd}(G - T)$. Adding an edge between

1. two even components: we lose two even components and obtain a new even component, so $\text{odd}(H - T) = \text{odd}(G - T)$.
2. two odd components: we lose two odd components and obtain a new even component, so $\text{odd}(H - T) = \text{odd}(G - T) - 2$.
3. an odd component and an even component: we lose an odd component and an even component, and obtain a new odd component, so $\text{odd}(H - T) = \text{odd}(G - T)$.

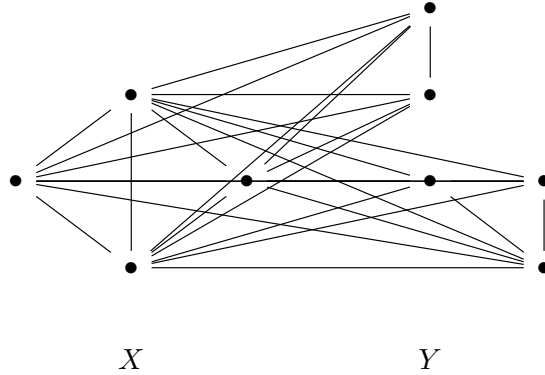
Hence $\text{odd}(H - T) \leq \text{odd}(G - T) \leq |T|$, as desired. $\square$

Definition 2.31. A graph G is Type-0 if $V(G)$ has a partition $X \cup Y$, where

$$X = \{x \in V(G) : xy \in E(G), \forall y \neq x, y \in V(G)\}$$

and every component of $Y = V(G) \setminus X$ is a complete graph.

Example 2.32. We have



Lemma 2.33. Let G be a graph of Type-0. If $\text{odd}(G - T) \leq |T|$ for all $T \subseteq V(G)$, then G has a perfect matching.

Proof. When $T = \emptyset$, we find $\text{odd}(G) = 0$, so $|V(G)|$ is even. When $T = X$, we find $\text{odd}(G - X) \leq |X|$. So there are at most $|X|$ odd complete components in $G - X$. Form a matching M by matching one vertex from each odd component of $G - X$ to a vertex in X . Match all remaining vertices in each component C of $G - X$ within C . Match remaining vertices in X within X (we can do it since we proved that $|V(G)|$ is even, and take out a vertex in the odd component and a vertex in X deducts 2 vertices, so the remaining vertices in X have an even number). $\square$

Theorem 2.34 (Tutte). A graph G has a perfect matching if and only if

$$\text{odd}(G - T) \leq |T| \text{ for all } T \subseteq V(G) \quad (**).$$

Proof. ($\implies$) observed already.

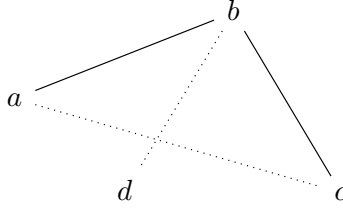
($\impliedby$) Assume G has property (**). Suppose, for contradiction, that G has no perfect matching. Let H be a graph such that

- $V(H) = V(G)$,
- $E(G) \subseteq E(H)$,
- H has no perfect matching,
- $H \cup \{ab\}$ for any $a, b \in V(H)$ where $ab \notin E(H)$ has a perfect matching. ($\dagger$)

Recall that G satisfies (**) would implies $|V(G)|$ is even. We can form H from G by adding edges until the last property holds. Then by the first lemma, H satisfies (**).

Claim: H is Type-0.

Claim. Let $X = \{x \in V(H) : xz \in E(H) \text{ for every } z \in V(H) \setminus \{x\} \text{ (possibly } \emptyset)\}$. Suppose C is a component of $H - X$ that is not complete, then $|V(C)| \geq 3$. We want to show that there exists vertices $a, b, c \in V(C)$, and $d \in V(H) \setminus \{a, b, c\}$ such that $ab, bc \in E(H)$, $ac \notin E(H)$, and $bd \notin E(H)$.



To see this, since C is not complete, we can find vertices a and y in C such that $ay \notin E(H)$. Let P be the shortest path from a to y . Let b and c be the second and third vertices on P , then $ac \notin E(H)$ because P is shortest. There exists d not adjacent to b because otherwise b is adjacent to every other vertex, then $b \in X$, but we choose $b \in V(G - X)$. By ($\dagger$), the graph $H \cup \{ac\}$ has a perfect matching M_1 , and $H \cup \{bd\}$ has a perfect matching M_2 .

Now, let $J = (V(H), M_1 \cup M_2)$. Then every component of G is

- a single edge in $M_1 \cap M_2$,
- an alternating cycle.

(no alternating path since M_1, M_2 both saturates $V(H)$). Since $ac \notin M_2$, it must be in an alternating cycle component U of J . If bd is not in U , then $M_1 \triangle E(U)$ is a perfect matching of H , contradicts to ($\dagger$). So we may assume $bd \in E(U)$. By symmetry of a, c , we may assume there is a path Q in U from d to a that does not contain b, c . Then the matching $M_2 \triangle U'$ is a perfect matching of H , where U' is the cycle $Q \cup \{ab, bd\}$, contradicts to ($\dagger$) of H . Therefore, every component C of $H - X$ must be complete, and therefore H is Type-0. $\square$

By the last lemma, we conclude that G has a perfect matching. $\square$

2.5 Applying Tutte's Theorem

Lemma 2.35. Let G be a graph with $|V(G)|$ even. Then for every $T \subseteq V(G)$, we have

$$\text{odd}(G - T) \equiv |T| \pmod{2}$$

Proof. Since $|V(G)| \equiv 0 \pmod{2}$, we find

$$|T| + |V(G - T)| = |V(G)| \equiv 0 \pmod{2}$$

$$\implies |T| \equiv |V(G - T)| \pmod{2}$$

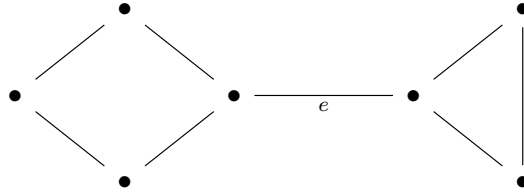
But,

$$\begin{aligned} |V(G - T)| &= \sum_{C \text{ odd}} |V(C)| + \sum_{C \text{ even}} |V(C)| \\ \implies |V(G - T)| &\equiv \sum_{C \text{ odd}} 1 + 0 = \text{odd}(G - T) \pmod{2} \end{aligned}$$

Therefore, $|T| \equiv |V(G - T)| \equiv \text{odd}(G - T) \pmod{2}$, as desired. $\square$

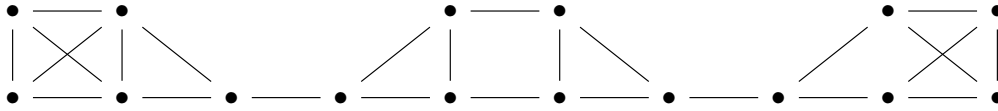
Definition 2.36. Recall from Math 239 that a bridge in a connected graph G is an edge e such that $G \setminus e$ is disconnected.

Example 2.37. e is the bridge for the graph below



Theorem 2.38 (Peterson). Let G be a connected cubic graph with at most two bridges. Then G has a perfect matching.

Example 2.39.



Proof. Note $|V(G)|$ is even, since it is cubic (Handshake lemma). Suppose that G has no perfect matching, then by Tutte's Theorem, there exists $T \subseteq V(G)$ with $\text{odd}(G - T) > |T|$. By the previous lemma, $\text{odd}(G - T) \geq |T| + 2$. (Notice that $T \neq \emptyset$ here).

Claim: For each odd component C of $G - T$, the number m_C of edges joining C to T is odd.

Claim. We have

$$3|V(C)| = \sum_{v \in V(C)} d_G(v) = 2|E(C)| + m_C$$

Therefore, m_C is odd since $|V(C)|$ is odd. $\square$

For at most two components C , we have $m_C = 1$, because such a single edge is a bridge of G . So the number of edges joining odd components of $G - T$ to T is at least $3(\text{odd}(G - T) - 2) + 2$. Also,

$$3(\text{odd}(G - T) - 2) + 2 \geq 3(|T| + 2 - 2) + 2 = 3|T| + 2$$

But since the graph is cubic, this is not possible as T can take at most $3|T|$ edges. $\square$

2.6 Defect Version of Tutte's Theorem

Theorem 2.40. Let G be a graph and let $d = \max_{T \subseteq V(G)} \{\text{odd}(G - T) - |T|\}$. Then G has a matching that saturates at least $|V(G)| - d$ vertices.

Proof. Note $d > 0$ since $\text{odd}(G - \emptyset) - |\emptyset| \geq 0$. If $d = 0$, then this is Tutte's theorem. So we assume $d \geq 1$. By the definition of d , for every $T \subseteq V(G)$, $\text{odd}(G - T) \leq |T| + d$. Construct a new graph H by adding a set A of d new vertices, and adding edges $\{ax : x \in V(H) \setminus \{a\}\}$ for each $a \in A$.

- $|V(H)|$ is even: for some $T_0 \subseteq V(G)$, $d = \text{odd}(G - T_0) - |T_0|$, so

$$|V(H)| = |V(G - T_0)| + |T_0| + d \equiv \text{odd}(G - T_0) - |T_0| + d \equiv 2d \equiv 0 \pmod{2}$$

- Let $\emptyset \neq S \subseteq V(H)$ where $A \not\subseteq S$. Then $\text{odd}(H - S) \leq 1$ since $H - S$ is connected. Hence $\text{odd}(H - S) \leq 1 \leq |S|$.
- Let $S = \emptyset$. Then $\text{odd}(H - S) = \text{odd}(H) = 0$ since H is connected and $|V(H)|$ is even.
- Let $S \subseteq V(H)$ where $A \subseteq S$.

Then $\text{odd}(H - S) = \text{odd}(G - (S \setminus A)) \leq |S \setminus A| + d = |S| - d + d = |S|$. Hence by Tutte's Theorem, H has a perfect matching M . At most d vertices of G can be incident to edges of M that are not edges of G (that is, their other endpoint is in A), so $M|_G$ saturates at least $|V(G)| - d$ vertices of G . $\square$

Theorem 2.41 (The Tutte-Berge Theorem). Let G be a graph and $d = \max_{T \subseteq V(G)} \{\text{odd}(G - T) - |T|\}$, then

$$\nu(G) = \frac{|V(G)| - d}{2}.$$

Proof. Recall from the previous proof that

$$|V(G)| - d \equiv |V(G)| + d \equiv 0 \pmod{2}$$

By the defect version of Tutte's theorem, we know

$$\nu(G) \geq \frac{|V(G)| - d}{2}$$

Let T_0 be such that $d = \text{odd}(G - T_0) - |T_0|$. Any matching in G can have an edge from at most $|T_0|$ odd components to T_0 . The remaining d odd components will each have an M -exposed vertex. So

$$|M| \leq \frac{|V(G)| - d}{2}.$$

$\square$

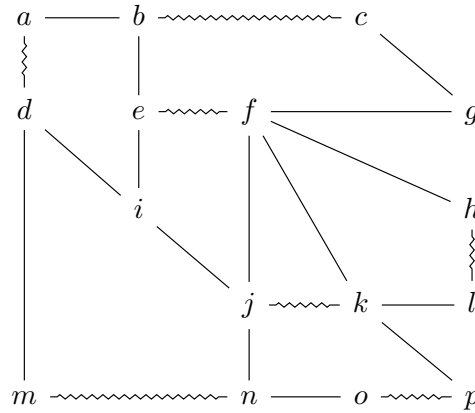
Edwards' Algorithm finds a maximum matching in a general graph. It is efficient.

2.7 Cycle Shrinking

Definition 2.42. Let G be a graph and let H be a matching in G . An odd cycle C of length $2k + 1$ is shrinkable with respect to M if

- $|M \cap E(C)| = k$, and
- C contains an M -exposed vertex.

Example 2.43. For the following graph



We have

$jkpon$: not shrinkable

$dmnji$: not shrinkable

$efkji$: shrinkable

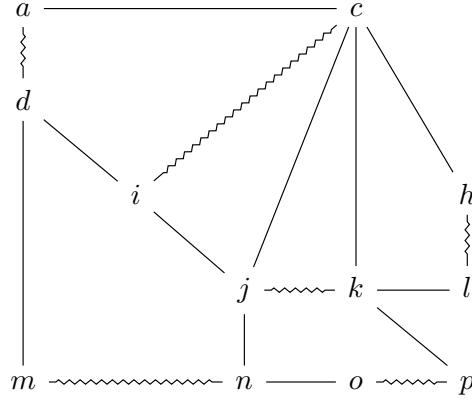
$bcgfe$: shrinkable

Lemma 2.44 (The Cycle Shrinking Lemma). Let M be a matching in a graph G . Suppose C is a shrinkable odd cycle with respect to M in G . Let G' be the graph obtained from G by contracting every edge of C (the operation of “cycle shrinking”). Then M is maximum in G if and only if the matching $M' = M \setminus E(C)$ is maximum in G' .

Proof. 略

□

Example 2.45. With cycle $C = bcgfe$, we get G' and M' :



2.8 M -alternating Trees

Definition 2.46. Let M be a matching in G . An M -alternating tree in G is a subgraph T of G where

- T is a tree
- T contains exactly one M -exposed vertex s called the root of T
- Each edge of T at an odd distance from s in T is an edge of M
- Each edge of T at an odd distance from s in T has degree 2 in T .

The vertices at odd distance from s in T are called the inner vertices of T . The set is denoted $I(T)$. All other vertices of T are called the outer vertices of T . The set is denoted $O(T)$.

Definition 2.47. An M -alternating forest in G is a subgraph F of G such that every component T of F is an M -alternating tree.

The sets $I(F)$ of inner vertices of F is $I(F) := \bigcup_{T \in F} I(T)$.

Similarly, the set $O(F)$ of outer vertices of F is $O(F) = \bigcup_{T \in F} O(T)$.

A maximal M -alternating forest is one that is not contained in a strictly larger M -alternating forest of G .

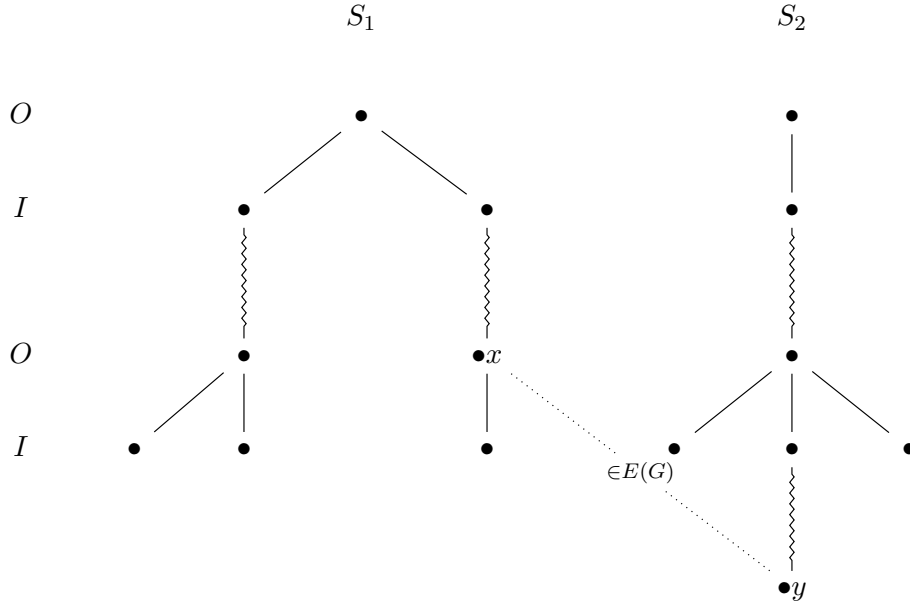
Lemma 2.48. Let F be a maximal M -alternating forest in G . If there is no edge of G that joins two outer vertices of F , then M is a maximum matching in G .

Proof. 略

□

Lemma 2.49. Let M be a matching in G , and let T_1 and T_2 be distinct components of an M -alternating forest F in G . Suppose $xy \in E(G)$ where $x \in O(T_1)$ and $y \in O(T_2)$. Then G contains an M -augmenting path.

Proof. Since the roots s_1, s_2 of T_1, T_2 respectively are both M -exposed, the unique path in $T_1 \cup T_2 \cup \{xy\}$ is an M -augmenting path (here we use also the definition of M -alternating forest).



□

Lemma 2.50. Let M be a matching in G . Let T be a component of an M -alternating forest F in G . Suppose there exists $xy \in E(G)$ such that $x, y \in O(T)$. Then there exists a matching $\bar{M}$ in G with $|\bar{M}| = |M|$ and a shrinkable odd cycle with respect to $\bar{M}$ in G .

Proof. Let C be the unique cycle in $T \cup \{xy\}$. Let P be the unique path from C to the root s of T in T . Set $\bar{M} = M \triangle E(P)$ (switch M in P). Then $|\bar{M}| = |M|$ since P is an (even) alternating path and C is shrinkable w.r.t. $\bar{M}$. □

2.9 Edwards' Algorithm for maximum matching

INPUT: A graph G .

OUTPUT: A maximum matching in G .

- Let M be a matching in G (e.g., $M = \emptyset$ or greedy).
- IF there are no M -exposed vertices, then we get a perfect matching, which is maximal. STOP and OUTPUT M . ELSE: Begin constructing an M -alternating forest F in G .
 - IF there exists $xy \in E(G)$ joining two outer vertices in the same component of F : replace M by $\bar{M}$, shrink the shrinkable odd cycle C given by the lemma to get a new graph G' . Update G to be G' , M to be $\bar{M} \setminus C$. Start over with this new G and M . (Back to step 2)
 - ELIF there exists $xy \in E(G)$ joining two outer vertices in distinct components of F : THEN replace M by $M \triangle E(P)$, where P is the M -augmenting path guaranteed by the first lemma. Expand all shrunk odd cycles and add as many edges of these cycles to

- M as possible. Then we get the original input graph G , and a larger matching. Start over with this G and the bigger matching M . (Back to step 1)
- ELSE: We complete the construction of a maximal M -alternating forest, with no edge of G joining the outer vertices of F . STOP. OUTPUT the matching in the original graph that led to this point (i.e., some cycles may have been shrunk).

2.10 Even Graphs

Definition 2.51. A graph G is said to be even if $d_G(v)$ is even for each $v \in V(G)$.

Definition 2.52. An Euler tour of a graph G is a sequence $v_0 e_1 v_1 e_2 v_2 \dots e_k v_k$ of vertices and edges of G such that

- $e_i = v_{i-1} v_i$ for each i
- $v_k = v_0$
- $e_i \neq e_j$ for all $i \neq j$
- $k = |E(G)|$ (that is, every edge of G appear exactly once).

Remark 2.53. If G has an Euler tour, then G is even.

Theorem 2.54. If G is even and connected, then it has an Euler tour.

Definition 2.55. For non-negative integer r , an r -factor in a graph G is an f -factor for the function $f : V(G) \rightarrow \{0, 1, 2\}$ defined by $f(v) = r$ for all $v \in V(G)$.

Theorem 2.56 (Peterson's 2-factor theorem). Let $d \geq 2$ be even. Then every d -regular graph G has a 2-factor.

Proof. We may assume G is connected, otherwise just look at each component. Then G has an Euler tour Q . Define the bipartite graph B as follows:

- $V(B) = V_0 \cup V_1$, where V_0 and V_1 are disjoint copies of $V(G)$.
- $E(B) = \{v_0 w_1 : vw \in E(G), v_0 \text{ is a copy of } v \text{ in } V_0, w_1 \text{ is a copy of } w \text{ in } V_1, \text{ and } vw \text{ is traversed in the direction } v \rightarrow w \text{ in } Q\}$

Then every vertex of B is of degree $\frac{d}{2}$ since each $v \in V$ has exactly $\frac{d}{2}$ out-edges in Q (which is $d_B(v_0)$), and similarly $d_B(w_0)$. Then B has a perfect matching M since it is regular of positive degree. Then M consists of exactly one in-edge and one out-edge for every vertex $v \in V(G)$. So M defines a 2-regular spanning subgraph of G , i.e., a 2-factor in G . $\square$

2.11 Vector Spaces Associated with Graphs

Recall: The principal proposition of a vector space W over a field $\mathbb{F}$:

- W contains a zero vector,
- any two $u, v \in W$, $u + v \in W$,

- any $v \in W$, $c \in \mathbb{F}$, $cv \in W$.

If $U \subseteq W$ and U is also a vector space over $\mathbb{F}$, we say U is a subspace of W .

Definition 2.57. Let G be a graph, and let S be a spanning subgraph of G (that is, $V(S) = V(G)$). The characteristic vector w_s of S is the $(0, 1)$ -vector with coordinates indexed by $E(G)$, defined by

$$(w_s)_i = \begin{cases} 1 & \text{if } e_i \in E(S) \\ 0 & \text{ow} \end{cases}$$

2.12 The Flow Space

Definition 2.58. Let G be a graph. The flow space $W(G)$ of G is the set of all characteristic vectors of even spanning subgraphs of G . This is a vector space over the field $\mathbb{Z}_2$:

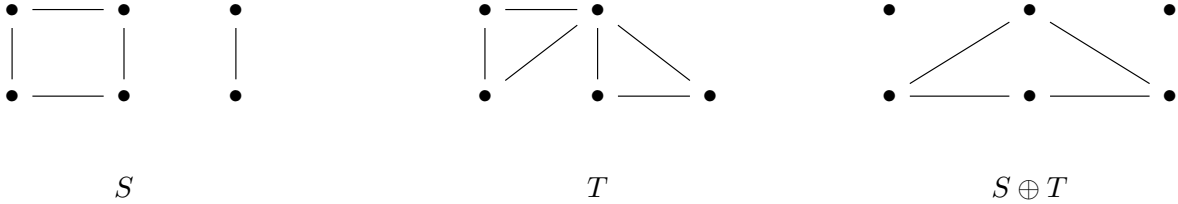
- $W(G)$ contains a zero factor because the spanning subgraph of G with 0 edges is even.
- addition in $w(G)$ is the symmetric difference. $w_S + w_T$ is the characteristic vector of the spanning subgraph $S \oplus T$ which has edge set $E(S) \cup E(T) \setminus E(S) \cap E(T)$. Then

$$d_{S \oplus T}(v) = d_S(v) + d_T(v) - 2d_{S \cap T}(v)$$

which is even. Therefore $w_S + w_T \in W(G)$.

- scalar multiplication is trivial.

Example 2.59.



Theorem 2.60. Let G be a graph and let H be an even spanning subgraph of G , then there exist cycles $C_1, \dots, C_k$ in G such that

$$E(H) = \bigsqcup_{i=1}^k E(C_i)$$

Proof. We will prove it by induction on $|E(H)|$.

Base case: If $|E(H)| = 0$, then take $k = 0$.

IH: Assume $|E(H)| > 0$, for any even spanning subgraph J of G , with $|E(J)| < |E(H)|$, we can find a set of disjoint cycles whose union gives J .

For some component H_1 of H , there exists a cycle in H_1 since $d_{H_1}(v) \geq 2$ for all $v \in V(H_1)$. Call this cycle C_1 . Then the spanning subgraph J defined by

$$E(J) = E(H) \setminus E(C_1)$$

Since every vertex in C_1 has degree 2, so

$$d_J(v) = \begin{cases} d_H(v), & v \notin C_1 \\ d_H(v) - 2, & v \in C_1 \end{cases}$$

which remains to be even. So J is an even spanning subgraph. By IH, there exists $C_2, \dots, C_k$ in J such that $E(J) = \bigsqcup_{i=1}^k E(C_i)$, hence

$$E(H) = \bigsqcup_{i=1}^k E(C_i).$$

□

Remark 2.61. This implies $W(G)$ is spanned by the set of char vectors of the cycles in G .

So now let's find a basis for $W(G)$.

Definition 2.62. Let G be a connected graph and T be a spanning tree on G . The unique cycle C_e in $T \cup \{e\}$ is called the fundamental cycle of e with respect to T , for each $e \in E(G) \setminus E(T)$.

Theorem 2.63. Let G be a connected graph and let T be a spanning tree of G . Then

$$B_T = \{w_C : C \in F(T)\}$$

is a basis for the flow space $W(G)$, where $F(T)$ is the set of fundamental cycles with respect to T .

Proof. First, we show B_T is linearly independent. Let $F(T) = \{C_e : e \in E(G) \setminus E(T)\}$. Suppose there are some $\lambda_e \in \{0, 1\}$ such that

$$\sum_{e \in E(G) \setminus E(T)} \lambda_e w_{C_e} = 0$$

For each $e \in E(G) \setminus E(T)$, the number of w_{C_e} that have a 1 in the e^{th} coordinate is 1, namely w_{C_e} . Hence $\lambda_e = 0$, as desired.

Now we want to show B_T spans $W(G)$: let H be an even spanning subgraph of G .

Claim: $w_H = \sum_{e \in E(H) \setminus E(T)} w_{C_e}$.

Note that this is the char vector of

$$\bigoplus_{e \in E(H) \setminus E(T)} C_e := J_H$$

We would like $J_H = H$, that is, $J_H \oplus H = \emptyset$. Each $e \in E(H) \setminus E(T)$ is an edge of J_H , since C_e is the unique fundamental cycle containing e , so $J_H \oplus H$ does not contain e . So $J_H \oplus H$ is a subgraph of T . If $J_H \oplus H$ has any edges, then it has a vertex of degree 1, so $J_H \oplus H$ is not even, a contradiction. Hence $J_H \oplus H$ is empty. □

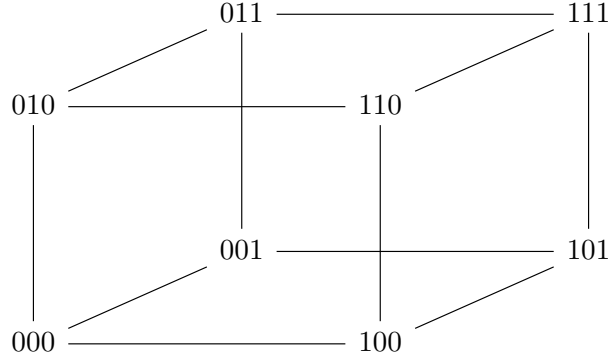
Corollary 2.64. For any connected graph G with n vertices and m edges, $\dim(w(G)) = m - n + 1$

Proof. Any spanning tree of G has $n - 1$ edges, so $\dim(w(G)) = m - (n - 1) = m - n + 1$. □

2.13 Binary Codes

Definition 2.65. A binary code of length m is a subspace U of $\mathbb{Z}_2^m$.

Example 2.66. 000 and 111 form a binary code.



Definition 2.67. The minimum distance of U is the smallest t such that some $0 \neq u \in U$ has exactly t 1's.

Definition 2.68. The Hamming distance between $u, v \in U$ is the number of 1-coordinates in $u \neq v$.

Lemma 2.69. Let U be a binary code with minimum distance t , then if $u, v \in U$ are distinct, then they are at Hamming distance at least t .

Proof. Since U is the subspace, $u + v \in U$. □

Definition 2.70. The girth of a graph G is the length of a shortest cycle. We say G has ∞ girth if G is a forest.

Lemma 2.71. Let G be a connected graph with finite girth g . Then the flow space $W(G)$ is a binary code of length $m = |E(G)|$, with dimension $m - n + 1$, where $n = |V(G)|$, and minimum distance g .

Proof. We just need to check that the minimum distance is g . Every nonzero element w_H of $W(G)$ is the char vector of H , an even spanning subgraph, which contains a cycle C . Since C has length at least g (g is the girth of G), w_H has at least g 1's (equality holds since a shortest cycle gives an even spanning subgraph). □